

Republic of Yemen
University of Science and Technology
Faculty of Engineering and Computing
Department of Biomedical Engineering



**Design and Implementation of an Integrated
EOG-Based Human-Computer Interface for
Virtual Keyboard and Wheelchair Control**

Research prepared by:

1. Abdulaziz Khaled Abdulaziz Hatem
2. Ahmed Mohammed Ahmed Salem Al-Kadhi
3. Mohammed Ayman Ali Qasem
4. Khaled Abdulmajid Mohammed Farhan

Supervised by:

Dr. Nasr Kaid Ali AL-Audi

*(To the Faculty of Engineering and Computing, This Research is a Part of the
Requirements for Obtaining a Bachelor's Degree in Biomedical Engineering)*

2025 - 2026

الإهداء

الحمد لله الذي وهبنا الإرادة، وذلّل لنا سبل العلم، وأعاننا بفضلِهِ حتى بلغنا تمام هذا العمل..

نهدي ثمرة جهدنا وسهرنا:

إلى من كانت دعواتهم طوق نِجاةٍ لنا في لحظات التعب، وإلى العيون التي سهرت تترقب نجاحنا..
آبائنا وأمهاتنا؛ كنزنا الأعلى، والسند الذي لا يميل.

إلى القلوب التي استلهمنا من صمتها الفكرة، ومن صلابة إرادتها العزيمة.. أصحاب الهمم من مرضى التصلب الجانبي الضموري، ومتلازمة المنحسب، والشلل الرباعي. نهديكم هذا الجهد بيقين وحب، عسى أن يكون يداً حانية ترد لكم شيئاً من استقالاتكم، ونافذة أملٍ تكسر حاجز الصمت وتصلكم بالعالم.

إلى إخوتنا، دفء الأيام وملاذنا الآمن.. وإلى أصدقاء المسيرة ورفاق الدرب، الذين تقاسمنا معهم ثقل البدايات ولذة النهايات، وكانوا لنا خير سندٍ ورفقة.

وإلى كل قلبٍ آمن بنا، وساندنا ولو بكلمة تشجيع صادقة..

إليكُم جميعاً نهدي هذا العمل المتواضع.

الشكر والعرفان

الحمد لله رب العالمين، حمداً يليق بجلال وجهه وعظيم سلطانه، على ما أنعم به علينا من توفيق وهداية، وما وهبنا من قدرة وصبر حتى وصلنا إلى إتمام هذا البحث وتكليل مسيرتنا العلمية بهذا الإنجاز.

وعملاً بالهدي النبوي الشريف: «مَنْ لَا يَشْكُرُ النَّاسَ لَا يَشْكُرُ اللَّهَ»، يُسعدنا ويشرفنا أن نتقدم ببالغ الشكر وعظيم الامتنان إلى مشرف مشرونا الفاضل الدكتور نصر قايد علي العودي، الذي كان لنصائحه القيمة وتوجيهاته الأكاديمية السديدة ومتابعته الدؤوبة أبلغ الأثر في تذليل الصعاب وتوجيه مسار هذا العمل الهندسي نحو أهدافه المرجوة.

كما نتقدم بخالص الشكر والتقدير إلى جميع أساتذتنا الأفاضل في قسم الهندسة الحيوية الطبية وكلية الهندسة في جامعة العلوم والتكنولوجيا، الذين لم يدخروا جهداً في تزويدنا بالعلوم والمعارف وإرساء القواعد العلمية والهندسية المتينة طوال سنوات دراستنا الجامعية.

ونتوجه بالشكر إلى كل من قدم لنا يد العون والمساعدة، أو أسدى إلينا نصيحة تقنية، أو دعمنا بكلمة مشجعة كان لها أثر طيب في إنجاح هذا العمل.

نسأل الله العليّ القدير أن يبارك في هذا الجهد، وأن يجعله نافعاً وخالصاً لوجهه الكريم، وأن يجزيهم جميعاً عنا خير الجزاء.

ABSTRACT

This project presents an integrated assistive medical platform designed for patients with severe motor disabilities, such as quadriplegia, locked-in syndrome, ALS, and muscular dystrophy, who retain voluntary eye movement. The core innovation lies in unifying autonomous smart wheelchair mobility and virtual keyboard communication into a single system controlled exclusively via Electrooculography (EOG) signals. The technical architecture features a custom EOG acquisition circuit using an AD620 instrumentation amplifier, advanced digital processing in MATLAB (250 Hz sampling rate, 50 Hz notch filter), and an Arduino-based embedded controller with ultrasonic obstacle avoidance. Evaluation across five subjects demonstrated an overall system accuracy of 92% and a typing speed of 16 characters per minute (CPM), proving that the proposed low-cost platform provides a reliable, safe, and effective solution to restore patient independence.

المستخلص

يُقدم هذا المشروع تصميمًا هندسيًا لمنصة طبية مساعدة مدمجة تلبي احتياجات المرضى الذين يعانون من إعاقات حركية شديدة (مثل الشلل الرباعي، متلازمة الانحباس، التصلب الجانبي الضموري، ALS وضمور العضلات) مع احتفاظهم بحركة العين. يكمن الابتكار الأساسي في دمج وظيفتين أساسيتين؛ وهما التنقل المستقل عبر كرسي متحرك ذكي، والتواصل عبر لوحة مفاتيح افتراضية، داخل منصة موحدة تعتمد كلياً على إشارات حركة العين (Electrooculography) - (EOG). تعتمد المعمارية التقنية على دائرة تماثلية مخصصة لاكتساب الإشارة وتكبيرها باستخدام مكبر الأجهزة (AD620) متبوعة بمعالجة رقمية متقدمة عبر بيئة (MATLAB) بمعدل أخذ عينات 250 Hz ومرشح (Notch Hz 50). أثبتت الدراسات التقييمية التي شملت خمسة متطوعين كفاءة النظام؛ حيث حقق دقة إجمالية بلغت (92%) ومتوسط سرعة كتابة (16 حرفاً في الدقيقة)، مدعوماً بنظام أمان آلي عبر حساسات (Ultrasonic) مما يجعله حلاً عملياً ومنخفض التكلفة لتحسين استقلالية المرضى.

Contents

List of Tables	viii
List of Figures	x
List of Abbreviations	1
Chapter 1 Introduction	2
1.1 General Overview and Clinical Background	3
1.2 Problem Statement	3
1.3 Project Engineering Objectives	4
1.4 Report Organization	4
1.5 Project Execution Timeline	5
Chapter 2 Literature Review and Theoretical Background	7
2.1 Medical Background	8
2.1.1 Anatomy and Physiology of the Human Eye (Corneo-Retinal Dipole)	8
2.1.2 Biomechanics of Eye Movements and Blinks	8
2.1.3 Characteristics of the Electrooculography (EOG) Signal	9
2.1.4 Physiological Principles of Electrode Placement	10
2.1.5 Pathophysiology of Target Neuromuscular Disorders	10
2.1.6 Clinical Justification for EOG-Based Human-Machine Interfaces	11
2.2 Engineering Background	11
2.2.1 Principles of Bio-Signal Acquisition and Ambient Noise	11
2.2.2 Instrumentation Amplifiers and the Significance of CMRR	12
2.2.3 Active Analog Filters (Active Sallen-Key and Notch Filters)	13
2.2.4 Digital Signal Processing and Pattern Extraction	14
2.2.5 Motor Control and Pulse-Width Modulation (PWM)	15
2.2.6 Principle of Ultrasonic Distance Measurement	16

2.3	Literature Review	17
2.3.1	Scientific Comparison: BCI (EEG) vs. EOG	17
2.3.2	Biosignal-Based Smart Wheelchairs	18
2.3.3	Virtual Keyboards and Spellers	19
2.3.4	Research Gap	20
2.4	Advantages and Feasibility of the Proposed Technique	21
2.4.1	Functional Integration in a Unified Platform	22
2.4.2	Blink-Based Control and Reduced Muscular Fatigue	22
2.4.3	Arabic Language Support	22
2.4.4	Navigational Safety and Caregiver Override	22
2.4.5	Technical and Clinical Feasibility	23
Chapter 3	System Architecture and Block Diagram	24
3.1	Overall System Block Diagram & Operation Flowchart	25
3.2	Subsystem Architecture Breakdown	26
3.2.1	Subsystem 1: Biopotential Acquisition and Analog Front-End (AFE)	26
3.2.2	Subsystem 2: Digitization and Central Signal Processing Pipeline	26
3.2.3	Subsystem 3: System Mode Manager and Output Interfaces	27
3.2.4	Subsystem 4: Autonomous Robotic Wheelchair and Safety Override	27
3.3	System Flowcharts	28
3.3.1	EOG Signal Acquisition, Processing, and Validation (MATLAB)	29
3.3.2	Unified System Mode Switching and Routing Logic	29
3.3.3	Autonomous Obstacle Avoidance and Output Actuation Flowchart	30
Chapter 4	Circuit Simulation and Hardware Implementation	31
4.1	Hardware Components Specifications	32
4.1.1	Instrumentation Amplifier (AD620)	32
4.1.2	Dual Operational Amplifier (TL072)	32
4.1.3	Bluetooth SPP Module (HC-05)	33
4.1.4	Dual H-Bridge Motor Driver (L298N)	33
4.1.5	Geared DC Motors (TT Motors)	34
4.1.6	Ultrasonic Time-of-Flight Sensor (HC-SR04)	35
4.1.7	Power Supply System: Lithium-Ion Cells and BMS	35
4.2	Analog Circuit Simulation and Mathematical Derivations	36

4.2.1	Bipolar Power Supply Management	37
4.2.2	Multi-Stage Hardware Topology	37
4.2.3	Mathematical Derivations & Theoretical Circuit Calculations . .	38
4.3	Analog Front-End Realization and Rapid Prototyping	40
4.4	Embedded Control and Autonomous Safety Hardware	41
4.5	Mechanical Design, Actuation, and 3D Printed Chassis	42
4.6	Complete System Integration	43
Chapter 5	Results and Discussion	45
5.1	EOG Signal Acquisition and Processing Results	46
5.1.1	Subject Demographics and Testing Protocol	46
5.1.2	EOG Waveform Acquisition and Digital Processing Pipeline . . .	46
5.2	Virtual Keyboard Performance Results	48
5.3	Wheelchair Navigation and Safety System Results	49
5.3.1	Motor Actuation and Command Responsiveness	49
5.3.2	Autonomous Obstacle Avoidance and Safety Override	50
5.3.3	Auto-Rotation Escape Maneuver	50
5.4	System Integration and Mode Switching Performance	51
5.5	General Discussion and Interpretation of Results	52
5.5.1	Impact of Hybrid Signal Processing	52
5.5.2	Overcoming Physiological Challenges	52
5.5.3	Clinical Value of the Integrated Design	53
5.6	Statistical Evaluation and Detailed Performance Metrics	53
5.6.1	Wheelchair Mobility Subsystem Performance	53
5.6.2	Virtual Optical Speller Performance and Text Entry Testing . . .	54
5.6.3	Integrated Performance Synthesis and Neuromuscular Learning Curve	55
5.6.4	Comparative Performance Benchmarking with Cited Scopus Lit- erature	56
5.6.5	Error Analysis and Physiological Limitations	58
Chapter 6	Conclusion and Recommendations	60
6.1	Conclusions and Significant Discoveries	61
6.2	Practical and Clinical Applications	61

6.3 Recommendations for Future Studies	62
References	63

List of Tables

1.1	Two-Year Project Execution Timeline (Year 3 – Year 4)	6
2.1	Comparative Scientific Summary of Prior Literature vs. Proposed Integrated System	21
5.1	Detailed Performance Metrics for Wheelchair Navigation Mode ($N = 5$)	54
5.2	Granular Performance Metrics for Virtual Optical Speller Mode ($N = 5$) .	55
5.3	Quantitative Performance Benchmarking Across System Domains (Scopus-Indexed Studies >2020 vs. Proposed System)	57

List of Figures

2.1	Corneo-retinal dipole potential and signal variations during eye movements [2].	8
2.2	Comparison of EOG signal waveforms: (A) Spontaneous involuntary blink displaying a narrow peak and short duration (≈ 150 ms); (B) Intentional voluntary blink exhibiting a wider peak, higher voltage amplitude, and extended duration (> 400 ms).	9
2.3	Vertical EOG electrode placement [4].	10
2.4	Conceptual comparison between an ideal clean EOG blink transient and a raw bio-signal contaminated with 50 Hz power-line interference and baseline drift.	12
2.5	Functional architecture and R_G gain-setting connection of the AD620 In-Amp [5].	13
2.6	Ideal frequency response curves (Bode plots) for High-Pass ($f_c = 0.8$ Hz), Low-Pass ($f_c = 30$ Hz), and 50 Hz Notch active filters in EOG signal conditioning [1].	14
2.7	Conceptual digital thresholding scheme and temporal pulse width (T_{width}) feature extraction for EOG event classification.	15
2.8	Pulse-Width Modulation (PWM) pulse trains at various duty cycles (25%, 50%, and 75%) and their corresponding average output voltages.	16
2.9	Operating principle of ultrasonic Time-of-Flight (ToF) distance measurement showing acoustic pulse emission, obstacle reflection, and two-way travel path ($2d$).	17
3.1	Overall System Block Diagram illustrating the Input Stage, Processing Unit, Output Stage, and Power Management Unit.	25
3.2	Unified System Operation and Decision Logic Flowchart.	28
4.1	AD620 Instrumentation Amplifier IC.	32

4.2	TL072 JFET-Input Dual Operational Amplifier.	33
4.3	HC-05 Bluetooth Module for wireless telemetry.	33
4.4	L298N Dual H-Bridge Motor Driver.	34
4.5	TT Geared DC Motor (1 : 48 Ratio).	34
4.6	HC-SR04 Ultrasonic Distance Sensor.	35
4.7	Power supply system components comprising 18650 Li-ion cells and 3S 40A BMS protection board.	36
4.8	Complete schematic diagram of the multi-stage EOG analog front-end cir- cuit.	37
4.9	Sequential multi-stage processing cascade of the EOG analog front-end. .	38
4.10	The physical realization of the EOG analog front-end on a breadboard. This modular rapid-prototyping setup was specifically adopted to allow real-time calibration of the potentiometers and dynamic adjustment of fil- ter stages during clinical testing.	41
4.11	Rear view of the wheelchair prototype highlighting the rear HC-SR04 ul- trasonic sensor responsible for the 30 cm reverse safety braking threshold. .	42
4.12	3D-printed wheelchair prototype mechanical design, double-deck PLA chassis layout, and 4WD actuation assembly.	43
4.13	Complete experimental setup demonstrating real-time EOG signal acqui- sition, hardware conditioning on the breadboard, MATLAB digital pro- cessing on the host machine, and signal verification via oscilloscope. . . .	44
5.1	Experimental test setup.	46
5.2	Raw EOG waveform on oscilloscope.	47
5.3	Processed EOG signal in MATLAB.	47
5.4	Virtual keyboard main interface.	48
5.5	Character selection inside a chosen sector.	49
5.6	Ultrasonic Time-of-Flight (ToF) obstacle detection and proximity thresh- old monitoring.	50
5.7	User interface in Wheelchair Mode.	51

LIST OF ABBREVIATIONS

Abbreviation	Description
ADC	Analog-to-Digital Converter
BMS	Battery Management System
CMRR	Common-Mode Rejection Ratio
EEG	Electroencephalography
EMG	Electromyography
EOG	Electrooculography
FSM	Finite State Machine
GUI	Graphical User Interface
HPF	High-Pass Filter
LIS	Locked-in Syndrome
LPF	Low-Pass Filter
MCU	Microcontroller Unit
PCB	Printed Circuit Board
PWM	Pulse-Width Modulation

CHAPTER 1

INTRODUCTION

This chapter introduces the fundamental clinical background, problem statement, and engineering objectives of the proposed EOG-based assistive technology system. It contextualizes non-invasive Human-Machine Interfaces (HMIs) for patients with Locked-in Syndrome (LIS), outlines the specific project goals, presents the structure of the thesis report, and details the two-year engineering milestone execution timeline.

1.1 General Overview and Clinical Background

Severe neurodegenerative disorders and traumatic injuries—such as advanced Amyotrophic Lateral Sclerosis (ALS), Muscular Dystrophy, brainstem stroke, and high-level spinal cord injuries—frequently result in profound motor paralysis, often presenting as Locked-in Syndrome (LIS) or Quadriplegia. Patients in these clinical conditions suffer from a complete loss of voluntary muscle control throughout the body, depriving them of conventional speech and physical interaction with their environment. However, the cranial nerves controlling extraocular muscle motility remain physiologically intact and functionally operational in most affected individuals.

To address these severe functional deficits, non-invasive Human-Machine Interfaces (HMIs) based on electrooculography (EOG) have emerged as an effective assistive solution. EOG exploits the standing electrostatic potential maintained between the cornea and the retina to track eye displacement and voluntary eyelid blinks.

1.2 Problem Statement

Individuals with severe motor paralysis face extreme social isolation and total reliance on caregivers due to the lack of functional communication and mobility tools. Although various assistive technologies currently exist, they present major technical and practical limitations. First, invasive Brain-Computer Interfaces (BCIs) provide direct neural control but carry significant surgical risks, tissue infection hazards, and high financial costs. Second, Video-Oculography (VOG) camera-based eye-tracking alternatives are highly sensitive to ambient lighting variations, require continuous spatial calibration, and demand high computational processing power. Third, system fragmentation remains a major barrier, as most commercially available devices treat communication and physical locomotion as separate challenges, requiring independent and expensive standalone platforms for each function.

Consequently, there is a clear clinical and engineering need for a non-invasive, robust, and cost-effective HMI. Such a system must utilize simplified bio-potential acquisition hardware to deliver real-time, context-aware control over both digital text communication and automated mobility platforms simultaneously.

This project proposes the design and implementation of a comprehensive assistive technology system that integrates an automated virtual keyboard for text-based communi-

cation and an engineered prototype vehicle controlled via adaptive EOG signal processing. This integrated framework simultaneously addresses mobility and communication limitations within a single, unified platform.

1.3 Project Engineering Objectives

The primary objective of this project is to develop an integrated, cost-effective, non-invasive HMI that empowers individuals with LIS to achieve independent digital communication and basic physical mobility. To attain this overall goal, the following specific engineering objectives were established:

- 1. Bio-Potential Acquisition Hardware Design:** To design and optimize a high-gain, low-noise analog front-end circuit capable of reliably capturing microvolt-level corneo-retinal potentials while rejecting power-line interference and environmental electromagnetic noise.
- 2. Real-Time Signal Processing Framework:** To implement real-time digital filtering and thresholding algorithms within a hybrid MATLAB-Arduino processing architecture to accurately extract and classify the temporal characteristics of voluntary eye blinks.
- 3. Context-Aware Control Integration:** To engineer a deterministic Finite State Machine (FSM) that translates multi-blink sequences into context-dependent operational commands, allowing a single bio-potential channel to manage two distinct subsystems while enforcing safety protocols.
- 4. Virtual Optical Speller Development:** To construct an auto-scanning virtual keyboard featuring a Graphical User Interface (GUI) optimized to reduce cognitive load, enabling users to compose, edit, and clear text efficiently.
- 5. 3D-Printed Prototype Wheelchair Implementation:** To construct a fully functional 3D-printed robotic wheelchair prototype that receives wireless directional commands, serving as a functional scaled model to validate smart wheelchair navigation algorithms.

1.4 Report Organization

This thesis is organized into six main chapters structured logically to document the complete engineering lifecycle of the EOG-based assistive system:

- **Chapter 1 (Introduction):** Outlines the clinical background, problem statement, engineering objectives, thesis structure, and execution timeline.
- **Chapter 2 (Literature Review and Theoretical Background):** Reviews existing literature on biopotential acquisition, HMIs, virtual spellers, and eye-tracking systems while identifying current technical gaps.
- **Chapter 3 (System Architecture and Block Diagram):** Details the high-level architecture, functional block diagrams, operational mode-switching logic, and overall system flowcharts.
- **Chapter 4 (Circuit Simulation and Hardware Implementation):** Presents circuit simulations, mathematical design calculations, component selection (AD620, L298N), and software architecture.
- **Chapter 5 (Results and Discussion):** Evaluates experimental biopotential waveforms, obstacle avoidance response times, virtual speller accuracy, and statistical performance metrics.
- **Chapter 6 (Conclusion and Recommendations):** Synthesizes the core engineering findings, clinical applications, and provides strategic recommendations for future technical developments.

1.5 Project Execution Timeline

The development lifecycle of this project was systematically organized across two academic years (within the four-year Biomedical Engineering curriculum).

During Year 3, efforts focused on EOG signal acquisition, continuous hardware design, noise suppression, and circuit optimization until a stable signal output was achieved. Building upon this hardware foundation, Year 4 focused on software development, virtual keyboard GUI programming, finite state machine integration, 3D printing of the prototype vehicle, and wireless control integration. Table 1.1 summarizes the engineering milestones achieved across this timeline.

Table 1.1: Two-Year Project Execution Timeline (Year 3 – Year 4)

Stage	Engineering Task & Technical Focus	Academic Period
Phase 1	Clinical background research, initial EOG signal acquisition, and preliminary analog circuit prototyping	Year 3 – Semester 1
Phase 2	Hardware optimization, active filter tuning, and achieving stable signal acquisition	Year 3 – Semester 2
Phase 3	Virtual keyboard programming (GUI), FSM control logic implementation, and signal algorithm refinement	Year 4 – Semester 1
Phase 4	3D-printed vehicle construction, wireless control integration, system validation testing, and final documentation	Year 4 – Semester 2

In summary, this introductory chapter established the clinical motivation, engineering scope, and milestone timeline of the proposed EOG-guided assistive system. Having outlined the foundational objectives for vehicle mobility and virtual speller communication, Chapter 2 provides the comprehensive literature review, medical physiology of the corneal-retinal dipole, and the biopotential conditioning engineering principles essential for system realization.

CHAPTER 2

LITERATURE REVIEW AND THEORETICAL BACKGROUND

This chapter provides the theoretical and scientific foundation supporting the proposed integrated assistive system. It begins with the medical background necessary to understand the biophysical nature and origin of electrooculography (EOG) signals. It then discusses the fundamental engineering principles related to biomedical signal acquisition, digital signal processing, and physical control systems. Furthermore, the chapter presents a literature review of previous studies in the field of eye-tracking and biomedical assistive technologies. Finally, it concludes by highlighting the advantages of the adopted technical approach and demonstrating the technical feasibility of this integrated design.

2.1 Medical Background

2.1.1 Anatomy and Physiology of the Human Eye (Corneo-Retinal Dipole)

From an electrical perspective, the human eye functions as a biological dipole or bio-battery. This electrostatic property arises from the resting potential difference maintained between the cornea and the retina. The anterior segment (cornea) carries a positive electrical charge, whereas the posterior segment (retina) carries a negative charge. This charge distribution generates an internal standing potential ranging from 10 mV to 30 mV [1]. However, when measured at the skin surface, this potential experiences significant attenuation due to the impedance of surrounding biological tissues. As the eyeball rotates, the orientation and magnitude of this electrostatic field shift relative to fixed electrode placement sites on the face. Figure 2.1 illustrates this corneo-retinal dipole mechanism and the resulting voltage variation observed at the skin surface during horizontal eye movements.

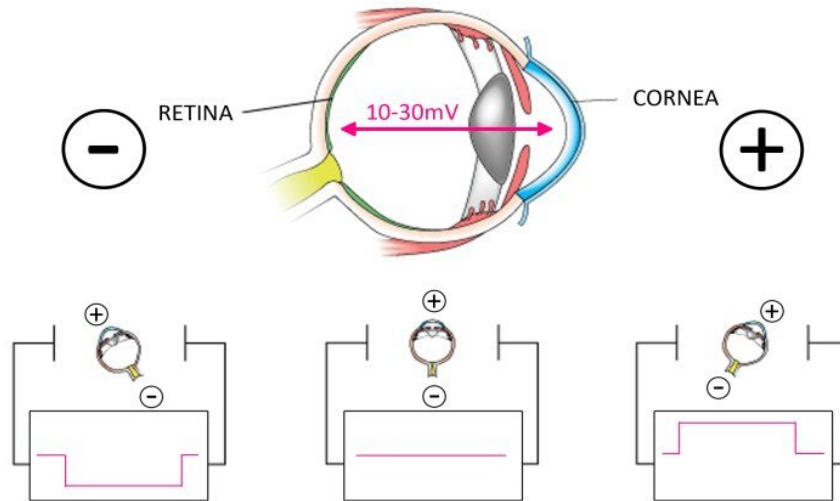


Figure 2.1: Corneo-retinal dipole potential and signal variations during eye movements [2].

2.1.2 Biomechanics of Eye Movements and Blinks

Human ocular dynamics consist primarily of rapid ballistic movements known as saccades and stationary resting intervals referred to as fixations. In addition, eyelid activity (blinking) can be physiologically categorized into involuntary and voluntary blinks. Involuntary blinks occur spontaneously to lubricate the corneal surface, whereas voluntary blinks are intentional, displaying significantly higher signal amplitudes and longer durations [3]. Distinguishing between these two physiological behaviors is essential for the system's control logic. The software algorithm relies on detecting intentional, strong blinks and

sustained fixations to generate mode-switching and selection commands while filtering out random involuntary blinks to prevent false system activation.

To distinguish between spontaneous eyelid activity and intentional control commands, the temporal and amplitude characteristics of the biopotential waveforms were analyzed. As illustrated in Figure 2.2(A), an involuntary blink produces a narrow, lower-amplitude biphasic spike with a short temporal footprint (≈ 150 ms) resulting from spontaneous corneal lubrication. Conversely, as depicted in Figure 2.2(B), an intentional voluntary blink yields a significantly wider peak with a higher voltage amplitude and an extended temporal duration (> 400 ms) caused by conscious eyelid muscle contraction. These amplitude and temporal width disparities form the physiological basis for real-time command classification.

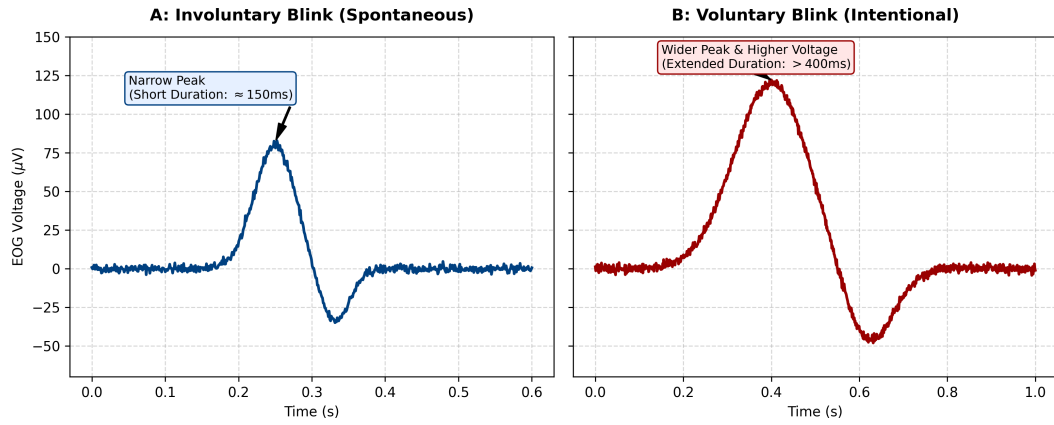


Figure 2.2: Comparison of EOG signal waveforms: (A) Spontaneous involuntary blink displaying a narrow peak and short duration (≈ 150 ms); (B) Intentional voluntary blink exhibiting a wider peak, higher voltage amplitude, and extended duration (> 400 ms).

2.1.3 Characteristics of the Electrooculography (EOG) Signal

The electrooculography (EOG) signal possesses specific biophysical characteristics that dictate the design requirements of the signal conditioning hardware. The amplitude of raw surface-measured EOG signals typically ranges from $10 \mu V$ to $5 mV$, necessitating high-precision differential instrumentation amplifiers. In terms of frequency distribution, the fundamental power spectrum of interest for ocular movement spans from 0.1 Hz to 30 Hz [1]. Based on these biophysical properties, a hardware low-pass filter with a cut-off frequency of 30 Hz was implemented to eliminate unwanted high-frequency noise. Furthermore, to isolate the DC offset component and shape the signal into sharp, well-defined pulses (spikes) corresponding to voluntary blinks, the cut-off frequency of the high-pass filter was set to 0.8 Hz.

2.1.4 Physiological Principles of Electrode Placement

Acquiring high-quality EOG signals requires proper anatomical positioning of surface electrodes. To capture the vertical channel (corresponding to vertical eye movements and eyelid blinks), electrodes are attached directly above and below one eye. For the horizontal channel, electrodes are placed at the lateral margins of both eyes, near the outer canthi. This configuration measures differential potential changes as the corneo-retinal dipole rotates toward or away from a given electrode, while a reference electrode positioned on the forehead provides a ground potential to cancel common-mode interference [1]. To illustrate the vertical channel electrode configuration described above, Figure 2.3 depicts the anatomical positioning of the supra-orbital and infra-orbital electrodes used in this project.

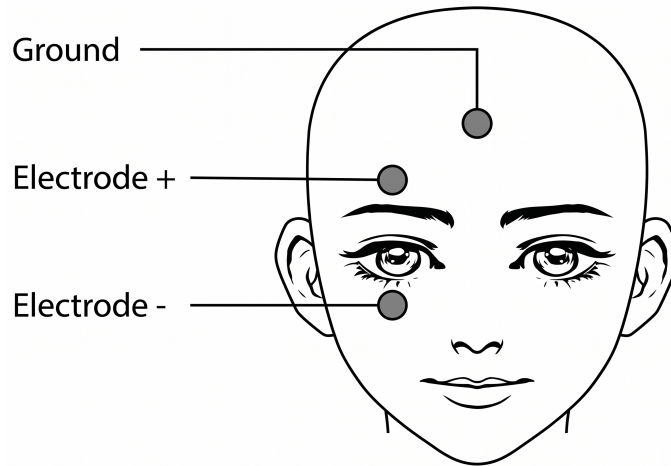


Figure 2.3: Vertical EOG electrode placement [4].

2.1.5 Pathophysiology of Target Neuromuscular Disorders

The developed assistive platform is designed for individuals suffering from severe neuromuscular impairment and loss of voluntary motor control. Key target pathologies include Amyotrophic Lateral Sclerosis (ALS), a progressive neurodegenerative disease; Quadriplegia, resulting in total limb paralysis; and Locked-in Syndrome (LIS), where cognitive awareness remains intact despite voluntary muscle loss. Furthermore, the system targets patients with Muscular Dystrophy (MD), a group of diseases causing progressive muscular weakness. In many stages of MD, the extraocular muscle control remains relatively preserved, making EOG an optimal, non-fatiguing physiological pathway for human-machine interaction [3].

2.1.6 Clinical Justification for EOG-Based Human-Machine Interfaces

The medical justification for selecting EOG signals as a control medium for these patient populations relies on an important anatomical and neurological principle: the cranial nerves responsible for extraocular muscle control—specifically the oculomotor nerve (CN III), trochlear nerve (CN IV), and abducens nerve (CN VI)—are typically spared during the progression of neurodegenerative conditions like ALS and severe paralysis [3]. Because ocular motility remains functionally preserved, voluntary eye movements represent a reliable, non-fatiguing physiological pathway through which severely impaired patients can interact with human-machine interfaces (HMIs) to regain independence in mobility and communication.

2.2 Engineering Background

Designing an EOG-guided 3D-printed prototype wheelchair and virtual speller system requires integrating bio-instrumentation, analog electronics, digital signal processing, embedded control, and distance sensing.

2.2.1 Principles of Bio-Signal Acquisition and Ambient Noise

Acquiring electrooculography (EOG) signals from the skin surface requires overcoming several physical and electrical challenges. First, high skin-electrode interface impedance significantly attenuates the microvolt-level biological signal. Second, the bio-potential is highly susceptible to ambient electromagnetic interference, primarily 50 Hz power line noise, which the human body picks up like an antenna and presents as a common-mode voltage across the measurement electrodes. Additionally, motion artifacts resulting from lead wire movement or adjacent electromyographic (EMG) muscle contractions contaminate the raw signal [1]. Overcoming these acquisition challenges requires designing a primary front-end amplifier characterized by ultra-high input impedance and superior common-mode rejection capability prior to any digital processing stage. To visualize the severity of these noise contamination sources, Figure 2.4 presents a conceptual comparison between a raw EOG signal corrupted by 50 Hz interference and baseline drift versus the ideal clean blink waveform that the signal conditioning chain aims to recover.

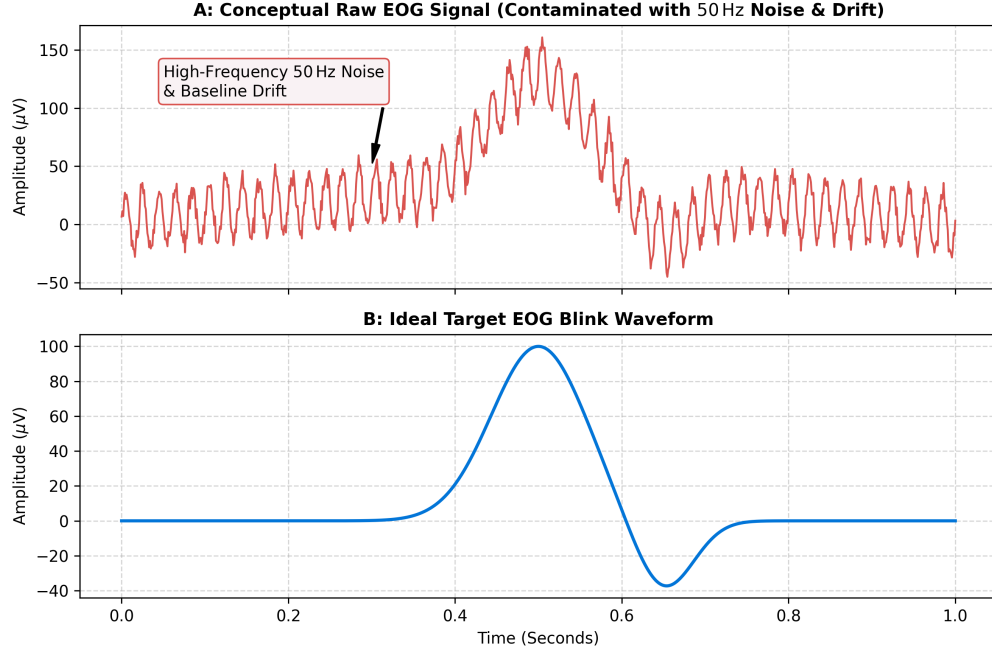


Figure 2.4: Conceptual comparison between an ideal clean EOG blink transient and a raw bio-signal contaminated with 50 Hz power-line interference and baseline drift.

2.2.2 Instrumentation Amplifiers and the Significance of CMRR

To isolate weak bio-potentials from severe ambient interference, an instrumentation amplifier (InAmp) is utilized in the primary acquisition stage rather than standard operational amplifiers (Op-Amps). An instrumentation amplifier operates on differential amplification principles; it amplifies the minute differential voltage existing between two measurement electrodes (representing ocular displacement) while virtually rejecting the common-mode potential present identically at both electrodes (such as 50 Hz power line noise) [1].

The efficiency of an amplifier in suppressing common-mode noise is mathematically defined by the Common-Mode Rejection Ratio (CMRR), expressed in decibels (dB) as:

$$\text{CMRR (dB)} = 20 \log_{10} \left(\frac{A_d}{A_{cm}} \right) \quad (2.1)$$

where A_d represents the differential voltage gain and A_{cm} represents the common-mode voltage gain.

In precision biomedical applications such as EOG acquisition, maintaining a high CMRR is essential to ensure signal integrity. This engineering requirement justifies selecting the AD620 integrated circuit (IC) for the primary pre-amplification stage in this project's hardware design. The AD620 provides an exceptionally high input impedance and a minimum CMRR of 100 dB [5].

In addition to noise rejection capabilities, the AD620 instrumentation amplifier allows precise control over the differential voltage gain (G) using a single external resistor (R_G), governed by the manufacturer's characteristic transfer equation [5]:

$$G = 1 + \frac{49.4 \text{ k}\Omega}{R_G} \quad (2.2)$$

This adjustable feature offers necessary engineering flexibility to achieve the target preliminary amplification for weak blink transients prior to passing the signal through subsequent active filtering stages. To illustrate this integrated architecture, Figure 2.5 depicts the internal functional schematic of the AD620 instrumentation amplifier and the external connection of the gain-setting resistor (R_G) across Pins 1 and 8 [5].

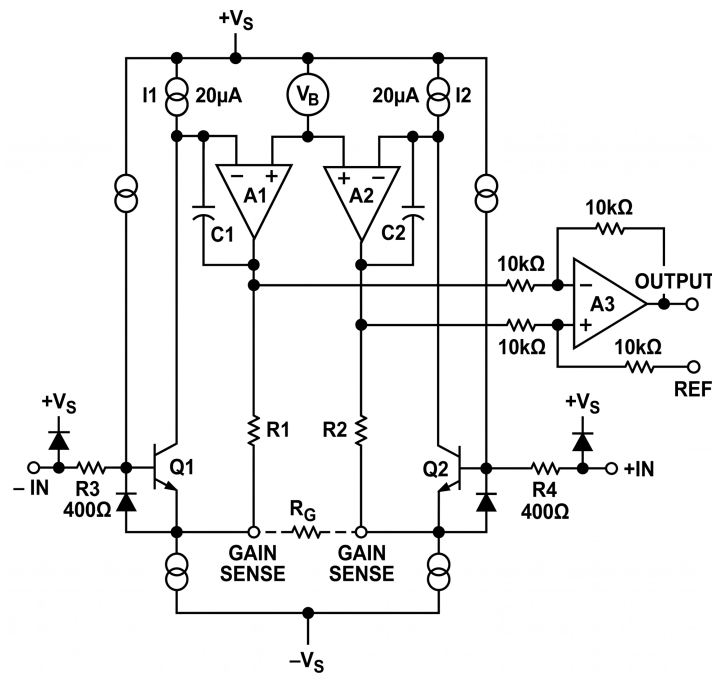


Figure 2.5: Functional architecture and R_G gain-setting connection of the AD620 InAmp [5].

2.2.3 Active Analog Filters (Active Sallen-Key and Notch Filters)

Active analog filters are critical for conditioning acquired bio-potentials prior to analog-to-digital conversion. Active topologies incorporating operational amplifiers are preferred in low-frequency biomedical applications because they eliminate the need for bulky inductors and offer low output impedance, which prevents inter-stage loading effects [1].

The Sallen-Key architecture is one of the most widely implemented topologies for designing second-order low-pass and high-pass filters. Assuming equal component values ($R_1 = R_2 = R$ and $C_1 = C_2 = C$), the cut-off frequency (f_c) is governed by:

$$f_c = \frac{1}{2\pi RC} \quad (2.3)$$

In biomedical signal processing, a high-pass filter (HPF) removes the DC offset caused by half-cell potentials at the skin-electrode interface, whereas a low-pass filter (LPF) restricts the signal bandwidth to satisfy sampling requirements and prevent aliasing. Furthermore, to eliminate 50 Hz or 60 Hz power-line interference, a notch filter (band-stop filter) is incorporated to sharply attenuate the specific power line frequency without distorting the primary biophysical spectrum. To visualize the spectral selectivity of these conditioning stages, Figure 2.6 illustrates the frequency response characteristics (Bode plots) of the high-pass ($f_c = 0.8$ Hz), low-pass ($f_c = 30$ Hz), and 50 Hz notch filters [1].

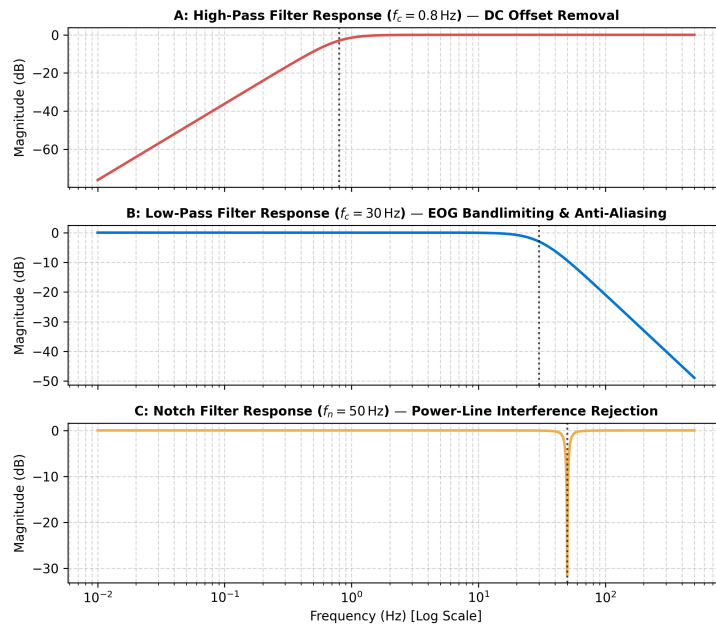


Figure 2.6: Ideal frequency response curves (Bode plots) for High-Pass ($f_c = 0.8$ Hz), Low-Pass ($f_c = 30$ Hz), and 50 Hz Notch active filters in EOG signal conditioning [1].

2.2.4 Digital Signal Processing and Pattern Extraction

Once digitized by an Analog-to-Digital Converter (ADC) at a sampling rate adhering to the Nyquist-Shannon sampling theorem, raw bio-signals undergo digital processing to enhance signal quality and extract feature metrics. Processing typically begins with digital

smoothing techniques, such as moving average filtering, to suppress high-frequency ripple components and highlight primary signal morphology.

Pattern extraction primarily relies on threshold detection algorithms. Mathematically, the instantaneous discrete signal sample $x(n)$ is compared against a predefined threshold value Th . When $x(n) > Th$, the event onset is logged. To differentiate between distinct physiological patterns (such as involuntary vs. intentional blinks), the algorithm measures the elapsed pulse width (T_{width}) during which the signal remains above the threshold. Based on this temporal width and event frequency, the signal is classified to generate real-time control commands for external hardware. To demonstrate this temporal feature extraction logic, Figure 2.7 depicts the threshold crossing criterion $x(n) > Th$ and the measurement of pulse width (T_{width}) used for pattern classification.

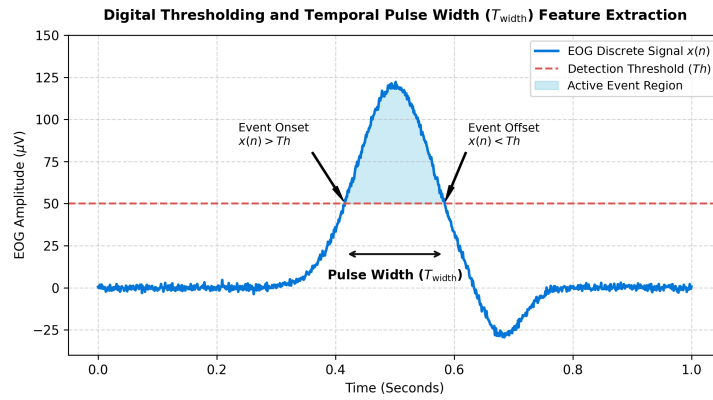


Figure 2.7: Conceptual digital thresholding scheme and temporal pulse width (T_{width}) feature extraction for EOG event classification.

2.2.5 Motor Control and Pulse-Width Modulation (PWM)

Guiding robotic and mobility systems requires precise control over both the rotational speed and direction of direct current (DC) motors. Because microcontrollers cannot directly deliver the high currents required by DC motors, motor driver circuits act as high-current electronic switches interfacing the controller to the motor power supply.

To regulate motor speed efficiently, Pulse-Width Modulation (PWM) is employed. PWM rapidly switches the applied voltage between ON and OFF states at a fixed switching frequency. The ratio of the ON duration to the total period, known as the duty cycle (D), determines the effective average output voltage (V_{out}) applied to the motor:

$$V_{\text{out}} = D \times V_{\text{in}} \quad (2.4)$$

where V_{in} represents the supply voltage. The duty cycle D is calculated as a percentage:

$$D = \frac{T_{on}}{T_{total}} \times 100\% \quad (2.5)$$

where T_{on} is the active pulse duration and T_{total} is the total period ($T_{on} + T_{off}$). This technique delivers linear speed control while minimizing thermal power dissipation. To illustrate the relationship between pulse active time and effective voltage, Figure 2.8 compares PWM pulse trains at different duty cycles and their resulting average output voltages.

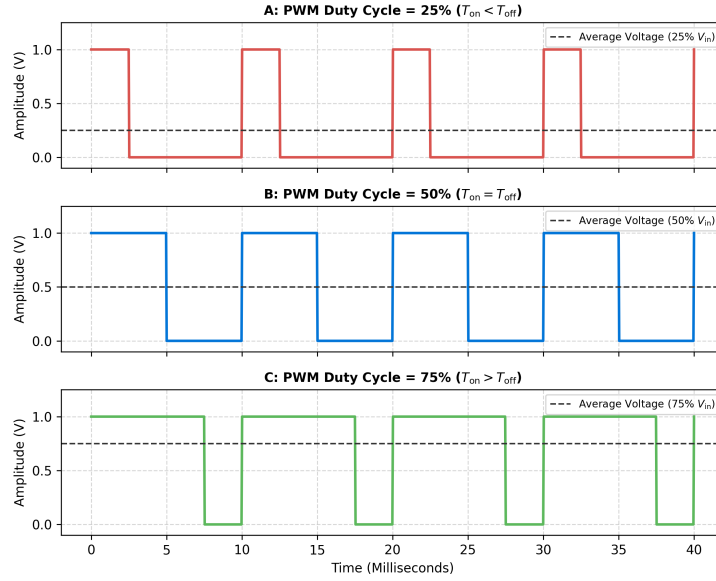


Figure 2.8: Pulse-Width Modulation (PWM) pulse trains at various duty cycles (25%, 50%, and 75%) and their corresponding average output voltages.

2.2.6 Principle of Ultrasonic Distance Measurement

To maintain navigation safety and prevent collisions in mobile platforms, ultrasonic sensors are implemented based on the Time-of-Flight (ToF) principle. The transmitter transducer emits ultrasonic acoustic pulses beyond human audible range, while the receiver transducer detects the echo reflected back from an obstacle.

The physical distance (d) between the sensor and the detected obstacle is calculated from the elapsed round-trip time (t) and the speed of sound in air ($v \approx 343$ m/s) using:

$$d = \frac{v \times t}{2} \quad (2.6)$$

The factor of 2 accounts for the two-way acoustic path (propagation to the obstacle and back). The computed distance feeds directly into safety control logic, executing imme-

diagnose automated intervention such as emergency stopping to prevent physical collision. To visualize this two-way acoustic path, Figure 2.9 illustrates the Time-of-Flight (ToF) measurement principle between the ultrasonic transducer and an obstacle.

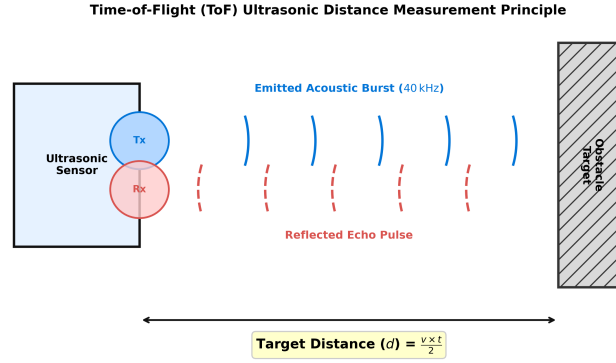


Figure 2.9: Operating principle of ultrasonic Time-of-Flight (ToF) distance measurement showing acoustic pulse emission, obstacle reflection, and two-way travel path ($2d$).

2.3 Literature Review

This section reviews existing literature on Human-Machine Interfaces (HMIs), smart wheelchairs, and assistive communication platforms. The goal is to analyze technical strengths and limitations, ultimately establishing the research gap addressed by the proposed integrated system.

2.3.1 Scientific Comparison: Brain-Computer Interfaces (EEG) vs. Electrooculography (EOG)

Electroencephalography (EEG) and electrooculography (EOG) represent two prominent bio-potentials utilized for assistive control interfaces [6]. Comparing both modalities based on prior literature reveals critical operational differences affecting their real-time feasibility:

- **Signal-to-Noise Ratio (SNR):** EOG signals exhibit a significantly higher SNR compared to EEG signals. Brain potentials are inherently microvolt-level, highly complex, and prone to severe interference from ambient noise and muscular activity [6].
- **Latency:** EEG-based systems typically suffer from relatively high latency due to the intensive signal processing required for pattern classification (e.g., P300 or Motor Imagery) [7]. Conversely, EOG signals provide faster response times and well-defined, consistent waveforms that are easily detected in real time.

- **Electrode Count:** EEG interfaces require high-density electrode caps covering the scalp, increasing hardware setup complexity and overall cost. In contrast, EOG systems require only a minimal number of facial surface electrodes placed around the eyes, offering superior user comfort [8].
- **Computational Complexity:** Decoding EEG signals demands advanced digital signal processing algorithms, imposing high computational loads. EOG signals possess distinct feature metrics (e.g., voluntary blink spikes) that can be processed on low-cost embedded microcontrollers [9].
- **Calibration Requirements:** Both modalities face operational calibration challenges. EOG signals are susceptible to baseline DC drift and ocular fatigue, requiring periodic baseline adjustments. Similarly, EEG systems demand extensive user training due to intra-subject signal non-stationarity.
- **Real-Time Performance and Clinical Comfort:** Voluntary eye movement provides an intuitive control modality that remains functional even in patients with severe paralysis. Although EOG control may sometimes be restricted to discrete command sets, it avoids the continuous cognitive strain associated with operating EEG-based interfaces [10].

Summary: Literature indicates that EOG is more suitable for real-time control of assistive wheelchairs and communication platforms due to its acquisition simplicity, cost-effectiveness, rapid response, and ability to deliver reliable control with minimal computational and physiological overhead [6, 10].

2.3.2 Biosignal-Based Smart Wheelchairs

Smart wheelchairs have evolved to enhance patient autonomy and reduce caregiver reliance. Prior studies have explored several bio-signal interaction modalities:

- **EEG-Based Systems:** Offered autonomous manual control based on Motor Imagery (MI) [11]. However, these systems exhibit degraded performance in dynamic environments due to user mental fatigue, command latency, and reliance on expensive instrumentation.
- **EOG-Based Systems:** Utilized ocular movements for direct directional control [8]. Prior studies demonstrate that EOG-based systems reduce setup time and deliver precise navigation commands [12].

- **Hybrid Systems (EEG/EOG):** Integrated EEG and EOG to overcome single-modality constraints; EOG artifacts (such as double blinks) were repurposed as rapid emergency stop commands, while EEG managed continuous directional steering [10, 13].

Engineering Limitations: Despite these advancements, many existing smart wheelchairs lack integrated autonomous obstacle avoidance mechanisms, exposing patients to collision risks and necessitating continuous caregiver supervision [14, 15]. Furthermore, high computational complexity hinders deployment on low-cost microcontrollers, while reliance on continuous mental focus imposes a heavy cognitive load on the user.

2.3.3 Virtual Keyboards and Spellers

To enable communication for severely motor-impaired individuals (e.g., LIS patients), various assistive interface modalities have been developed:

- **P300-Based Systems (EEG):** Rely on matrix spellers with flashing rows and columns, requiring intense visual attention from the user [16]. Although widely researched, P300 spellers exhibit low typing speeds, high visual fatigue, and low Information Transfer Rates (ITR).
- **Vision-Based / Eye-Tracking Systems:** Utilize camera-based video-oculography to track gaze direction [17]. While offering high spatial accuracy, these systems are highly sensitive to ambient lighting variations and require head-pose stability, limiting their operational effectiveness in dynamic outdoor environments.
- **EOG-Based Keyboards:** Demonstrate superior robustness under variable lighting conditions without requiring camera hardware [9, 18]. Character selection is achieved through directional eye movements and confirmed via voluntary blinks, achieving practical communication rates.

Arabic Support and Usability Limitations: The vast majority of existing EOG- and EEG-based communication platforms are tailored exclusively for Latin or Asian languages, leaving Arabic language support severely lacking [16, 12]. The Arabic language possesses distinct structural and grammatical characteristics that necessitate customized interaction interfaces. Recently, integrating Morse code logic with EOG signals has proven effective for Arabic character generation as a low-cost alternative [12]. Additionally, many EOG interfaces suffer from the “Midas Touch” problem—the inability to differentiate between spontaneous eye movements and intentional control commands—leading to frequent false triggers.

2.3.4 Research Gap

Critical analysis of existing literature reveals that current assistive technologies typically provide fragmented solutions: they either focus exclusively on mobility via smart wheelchairs or independently address virtual communication interfaces. Unification of both essential functionalities into a single platform is rarely addressed.

Furthermore, prior studies exhibit common technical limitations:

1. Absence of unified integration enabling seamless, dual-mode control over mobility and communication.
2. High implementation costs and computational complexity that prevent deployment on low-cost embedded systems.
3. A severe shortage of Arabic language support in bio-signal-based communication interfaces.
4. Lack of embedded autonomous navigation safety mechanisms (such as automatic obstacle avoidance) fully integrated with a single EOG control interface.

Based on this review, the primary research gap addressed by this project is formulated as follows:

Formulated Research Gap

Research Gap: The lack of a low-cost, unified assistive platform that effectively integrates safe autonomous wheelchair navigation (featuring automated obstacle avoidance) with Arabic virtual speller communication through a single, non-invasive EOG interaction interface.

To provide a structured synthesis of the literature, Table 2.1 presents a comparative scientific summary comparing prior assistive technologies against the proposed integrated system.

Table 2.1: Comparative Scientific Summary of Prior Literature vs. Proposed Integrated System

Modality	Key Advantages	Major Technical Limitations	Target Scope	Arabic & Safety
EEG / BCI [11, 7]	Direct neural decoding	Low SNR, high latency, complex electrode cap, mental fatigue	Single mode (Mobility/Speller)	Unsupported
VOG (Camera) [17]	High spatial accuracy	Sensitive to lighting, head posture drift, high compute load	Speller only	Rare / Limited
EOG Wheelchair [8, 13]	High SNR, low-cost hardware	Lacks real-time obstacle avoidance, no text speller	Navigation only	No safety override
EOG Speller [9, 16]	Robust in lighting changes	Midas-touch false triggers, non-Arabic script design	Speller only	Lacks native Arabic
Proposed System	High SNR, hybrid MATLAB-Arduino DSP, 3-electrode interface	Relies on voluntary blink temporal width classification	Dual-Mode (Wheelchair + Speller)	Full Arabic + Auto Obstacle Escape

2.4 Advantages and Feasibility of the Proposed Technique

The developed system presents an integrated engineering solution that directly addresses the technical and operational gaps identified in prior literature by combining 3D-printed prototype wheelchair navigation and an interactive virtual speller into a single platform. Rather than relying on fragmented, single-purpose assistive devices, the proposed framework adopts a holistic, patient-centered approach targeted at individuals with severe motor impairment, such as Quadriplegia and Locked-in Syndrome (LIS). The key engineering advantages and technical feasibility of this design are outlined below.

2.4.1 Functional Integration in a Unified Platform

To overcome the limitations of prior assistive systems that require separate devices for locomotion and communication, the proposed platform integrates both functions into a unified assistive medical system. Utilizing a single biological acquisition channel, raw electrooculography (EOG) signals are processed and routed to a central MATLAB Graphical User Interface (GUI). The user can seamlessly toggle between “Wheelchair Mode” and “Keyboard Mode” by executing a designated biological command sequence, specifically four consecutive voluntary blinks. This unified architecture eliminates the need for multiple hardware setups, reduces physiological and cognitive workload, and establishes a consolidated control environment that enhances patient autonomy.

2.4.2 Blink-Based Control and Reduced Muscular Fatigue

Unlike vision-based eye-tracking systems that suffer from lighting sensitivity or EEG interfaces that cause cognitive exhaustion, the proposed platform relies exclusively on voluntary blink dynamics (short blinks, long blinks, and distinct blink sequences). To minimize muscular fatigue and reduce the total number of required commands, a context-aware control algorithm was engineered. The algorithm dynamically reinterprets biological control signals depending on the active state of the vehicle or text interface. This state-dependent design provides an immediate, reliable system response with minimal physical exertion, aligning with the limited motor capabilities of LIS patients.

2.4.3 Arabic Language Support

Addressing the critical shortage of bio-signal-based communication interfaces tailored for Arabic speakers, the platform incorporates an innovative virtual keyboard featuring a circular layout divided into six sectors. The user interface automatically scans through the sectors sequentially, allowing the user to select the target sector and subsequent character via voluntary blinks. Beyond basic character input, the speller includes comprehensive text-editing features, such as space insertion, single character deletion, smart word deletion, and full display clearing. This dynamic interaction model bridges the language accessibility gap for Arabic-speaking patients.

2.4.4 Navigational Safety and Caregiver Override

Unlike conventional electric wheelchairs that place the full burden of collision avoidance on the user, the proposed 3D-printed smart wheelchair prototype incorporates an auto-

mated safety system. Front and rear ultrasonic distance sensors continuously monitor the immediate environment to detect obstacles and trigger emergency auto-stop protocols. Furthermore, to maximize operational safety, a mobile application provides a secondary wireless control channel (Caregiver Override) via Bluetooth. This override feature enables medical personnel or caregivers to instantly assume manual control of the wheelchair during emergencies or when patient fatigue occurs, balancing autonomous mobility with safety.

2.4.5 Technical and Clinical Feasibility

From an engineering perspective, the system demonstrates high feasibility by eliminating the need for expensive commercial bio-signal acquisition equipment. A custom, low-cost Analog Front-End (AFE) circuit was designed specifically to acquire microvolt-level EOG signals, utilizing an AD620 instrumentation amplifier alongside active analog and digital filtering (including a 50 Hz active notch filter for power line noise rejection). By employing reliable, off-the-shelf components—such as Arduino microcontrollers, an HC-05 Bluetooth module for wireless data transmission, and a rechargeable lithium-ion battery system managed by a Battery Management System (BMS)—the project achieves an optimal trade-off between performance accuracy and manufacturing cost. This low-cost modular design facilitates immediate clinical implementation and potential commercial scaling.

In conclusion, this chapter synthesized the physiological basis of EOG signals, biopotential conditioning electronics, and relevant academic literature. The established advantages of real-time responsiveness and unified hardware integration form the structural parameters for system design. Building upon these theoretical principles, Chapter 3 details the high-level functional block diagrams, operational mode switching logic, and system flowcharts.

CHAPTER 3

SYSTEM ARCHITECTURE AND BLOCK DIAGRAM

This chapter details the high-level architecture and operational behavioral logic of the EOG-guided assistive system. It presents the overall system block diagram detailing the input, processing, output, and power management units, followed by functional operational descriptions. Additionally, it outlines the software flowcharts for MATLAB EOG thresholding, unified mode switching, and Arduino obstacle avoidance. In strict accordance with system modeling standards, component-level hardware specifications are omitted from this chapter.

3.1 Overall System Block Diagram & Operation Flowchart

The complete system architecture is conceptualized as a multi-stage signal processing pipeline. Figure 3.1 illustrates the high-level block diagram comprising four primary functional stages:

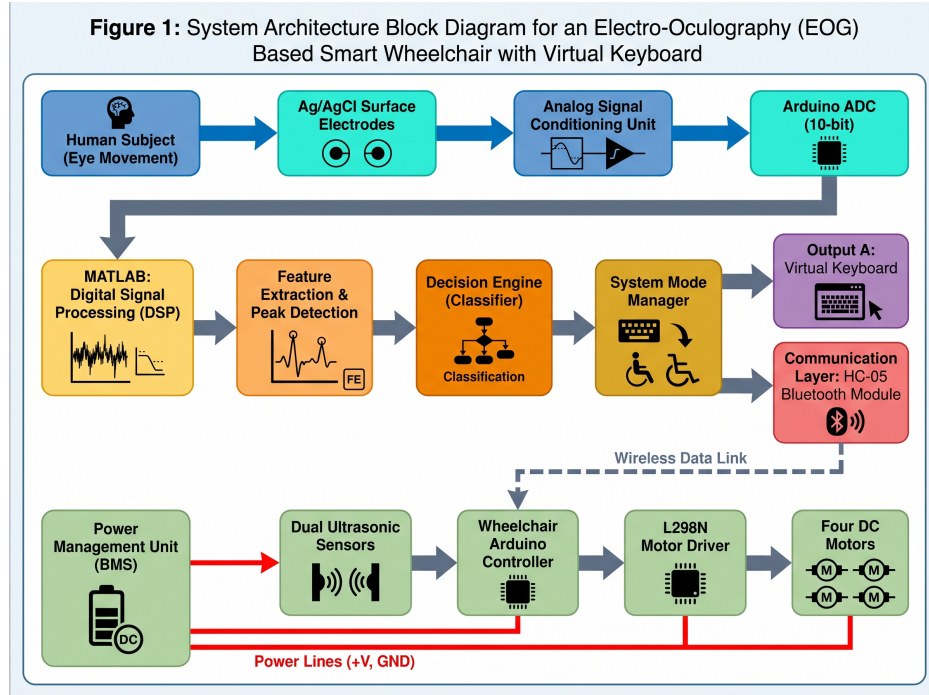


Figure 3.1: Overall System Block Diagram illustrating the Input Stage, Processing Unit, Output Stage, and Power Management Unit.

- 1. Input Stage:** Bio-potential skin electrodes, differential bio-amplifier, active analog filters, and DC level conditioning.
- 2. Processing Unit:** Primary signal processing unit (MATLAB environment) for blink detection and feature extraction, interconnected with a real-time embedded microcontroller unit.
- 3. Output Stage:** Dual output actuators consisting of the motorized 3D-printed prototype wheelchair drive mechanism and the Graphical User Interface (GUI) virtual speller.
- 4. Power Management Unit:** Regulated power distribution network supplying dual symmetric voltage rails for analog signal conditioning and logic power for digital modules.

3.2 Subsystem Architecture Breakdown

This section presents a comprehensive structural and functional breakdown of the four core subsystems comprising the integrated EOG-guided assistive platform, detailing the signal transformation pipeline from biopotential sensing to actuation and virtual communication.

3.2.1 Subsystem 1: Biopotential Acquisition and Analog Front-End (AFE)

- **Ag/AgCl Surface Electrodes:** Detects the microvolt-level potential differences generated by the standing corneo-retinal dipole during voluntary eye movements and blinks. Electrodes are placed in a vertical periorbital configuration with a forehead reference electrode.
- **Instrumentation Amplifier Stage (AD620):** Provides primary differential amplification for weak raw EOG potentials ($\approx 100 \mu\text{V} - 600 \mu\text{V}$). It leverages a high Common-Mode Rejection Ratio ($\text{CMRR} > 100 \text{ dB}$) and an ultra-high input impedance ($10^{10} \Omega$) to reject power-line noise and prevent skin-electrode loading effects.
- **Active Filtering Cascade (TL072 Op-Amps):** Consists of a second-order active Sallen-Key High-Pass Filter ($f_c = 0.8 \text{ Hz}$) to remove half-cell potential DC offsets and motion baseline drift, followed by a second-order Low-Pass Filter ($f_c = 30 \text{ Hz}$) to eliminate high-frequency electromyographic (EMG) muscle artifacts.
- **50 Hz Power-Line Notch Filter:** A dedicated active notch filter sharply attenuates 50 Hz ambient electromagnetic hum coupled from surrounding power infrastructure.
- **DC Level Shifter:** Adds a stable $+2.5 \text{ V}$ DC offset to the amplified bipolar signal ($\pm V$), converting it into a purely positive unipolar waveform ($0 \text{ V} - 5 \text{ V}$) fully compatible with the analog input limits of the microcontroller.

3.2.2 Subsystem 2: Digitization and Central Signal Processing Pipeline

- **Analog-to-Digital Conversion (ADC):** The conditioned analog waveform is sampled by a 10-bit embedded ADC at a continuous sampling frequency ($F_s = 250 \text{ Hz}$), quantizing the $0 - 5 \text{ V}$ input into digital values ($0 - 1023$).
- **Serial Data Streaming:** Digitized biopotential samples are packaged and transmitted to the host environment via UART serial communication at a baud rate of 115200 bps.
- **MATLAB DSP Engine:** Applies dynamic moving-average envelope filtering and secondary digital Butterworth notch filtering to yield a clean, smoothed waveform.

- **Feature Extraction & Peak Detection:** Computes instantaneous signal amplitude against defined threshold levels (V_{th}) and measures the pulse time duration (T_{width}) to reliably discriminate intentional control blinks from involuntary spontaneous blinks.

3.2.3 Subsystem 3: System Mode Manager and Output Interfaces

- **Finite State Machine (FSM) Mode Manager:** Serves as the central decision engine. It interprets blink counts and temporal sequences. Upon classifying a four-blink trigger sequence, the FSM toggles the active operational context between communication and mobility modes.
- **Virtual Optical Speller (Mode A):** When active, classified blink commands drive a circular 6-sector auto-scanning interface, enabling hands-free Arabic text entry, editing, and display management.
- **Wireless Telemetry (Mode B):** When Wheelchair Mode is active, classified directional commands are converted into control bytes and passed to an HC-05 Bluetooth master module for wireless transmission.

3.2.4 Subsystem 4: Autonomous Robotic Wheelchair and Safety Override

- **Embedded Control Receiver:** An HC-05 slave receiver mounted on the prototype wheelchair receives the wireless control frames and forwards them to the navigation microcontroller.
- **Ultrasonic Safety Array (HC-SR04):** Continuously polls front (40 cm threshold) and rear (30 cm threshold) distances using Time-of-Flight (ToF) acoustic measurements.
- **Autonomous Safety Override:** If an environmental obstacle breaches the safety perimeter, an immediate hardware interrupt suspends active user EOG drive commands, arresting motor power and triggering an automated escape rotation.
- **H-Bridge Power Stage & Actuation (L298N & DC Motors):** Drives four geared DC motors configured in a differential steering layout. The driver applies Pulse-Width Modulation (PWM) signals to ensure smooth acceleration, linear speed regulation, and zero-radius turns.

3.3 System Flowcharts

System software behavior is governed by the master system operation flowchart illustrated in Figure 3.2, alongside three primary sub-flowcharts executing concurrently across the host processing unit and embedded microcontroller.

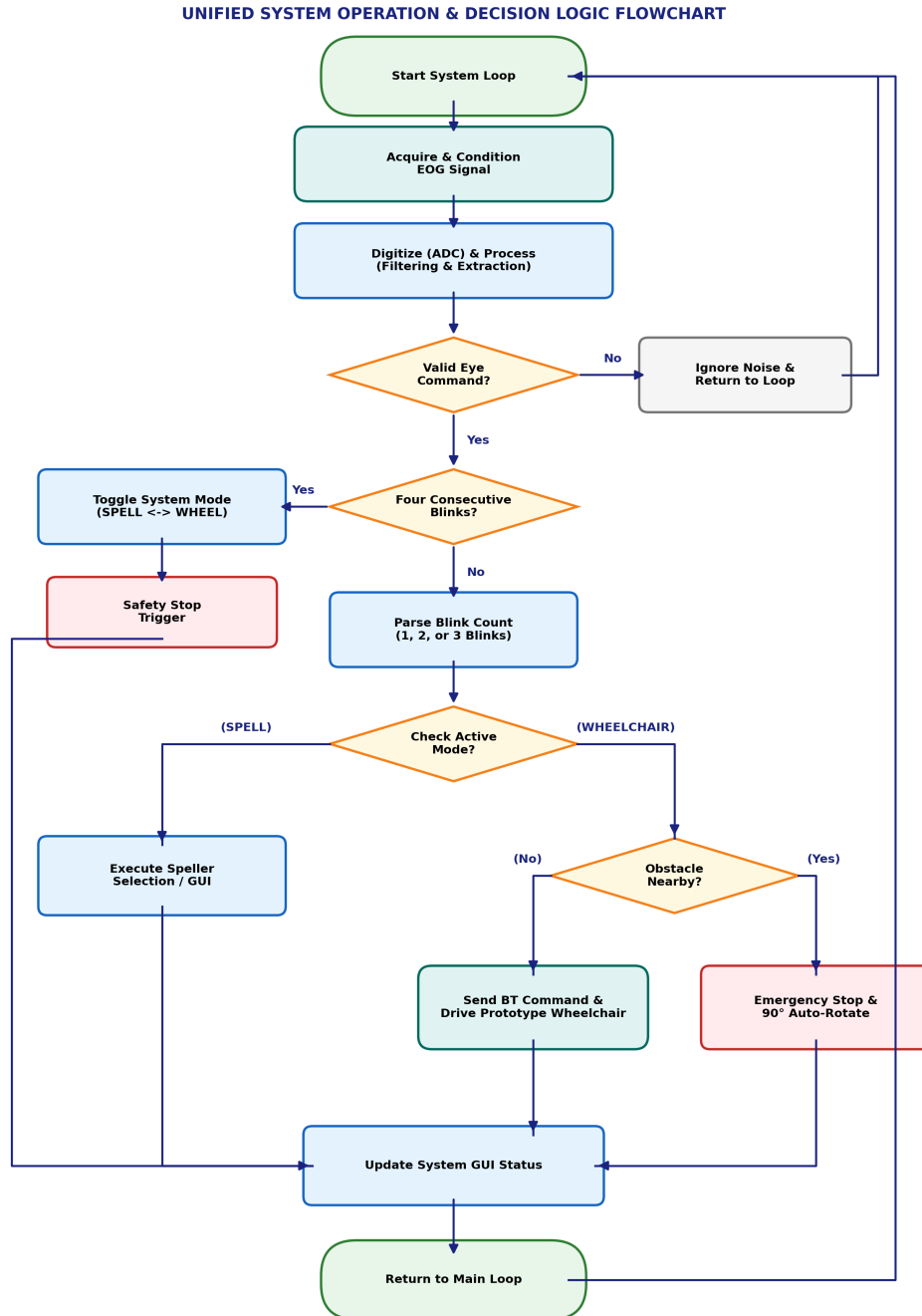


Figure 3.2: Unified System Operation and Decision Logic Flowchart.

3.3.1 EOG Signal Acquisition, Processing, and Validation (MATLAB)

The master execution loop initiates with the continuous acquisition of ocular micro-biopotentials via surface Ag/AgCl electrodes (*Start System Loop*). The raw physiological signal undergoes primary analog conditioning—utilizing the AD620 instrumentation amplifier and TL072 active filters—before being digitized by the microcontroller’s ADC at a locked frequency of 250 Hz (*Acquire & Condition EOG Signal*).

Upon transmission to the MATLAB environment, the digital signal is refined using a 50 Hz Notch filter and a moving-average algorithm to eliminate power-line hum and baseline noise (*Digitize & Process*). Subsequently, the algorithm evaluates the waveform against predefined amplitude thresholds (*Valid Eye Command?*). If the signal fails to meet the strict criteria (e.g., spontaneous blinks or motion artifacts), the system classifies it as a false trigger, ignoring the noise and resetting the cycle (*Ignore Noise & Return to Loop*). Only valid, intentional blink transients are permitted to proceed to the classification engine.

3.3.2 Unified System Mode Switching and Routing Logic

Once an intentional ocular command is validated, the Decision Engine determines the system’s operational path based on the temporal blink sequence. The logic first evaluates whether a high-level toggle sequence has been executed (*Four Consecutive Blinks?*).

If exactly four blinks are detected (*Yes*), the system executes a mode transition (*Toggle System Mode: SPELL ↔ WHEEL*). Crucially, to ensure patient safety during this macro-state transition, a *Safety Stop Trigger* is simultaneously activated, automatically placing the wheelchair into a STANDBY state to prevent sudden, uncommanded physical movement.

If the sequence contains fewer than four blinks (*No*), the logic progresses to evaluate the specific gesture (*Parse Blink Count: 1, 2, or 3 Blinks*) and routes the command based on the currently active macro-state (*Check Active Mode?*):

- **Virtual Speller Mode (SPELL):** The system routes the command to the GUI (*Execute Speller Selection*). The parsed blink count dictates the typing action: 1 blink executes sector/character selection, 2 blinks trigger a space or smart backspace, and 3 blinks clear the text display.
- **Wheelchair Mode (WHEELCHAIR):** The parsed directional command is routed to the physical navigation and safety layer.

3.3.3 Autonomous Obstacle Avoidance and Output Actuation Flowchart

When operating in Wheelchair Mode, user commands are not executed blindly. Before any directional command is translated into physical actuation, the decision logic queries the embedded ultrasonic safety layer (*Obstacle Nearby?*).

- **Collision Imminent (Yes):** If an environmental obstacle breaches the strict proximity thresholds, the user's directional intent is immediately overridden. The system executes a hardware-level intervention (*Emergency Stop & 90° Auto-Rotate*), autonomously halting the vehicle and rotating it until a clear path is verified.
- **Clear Path (No):** If the safety perimeters are clear, the validated movement command is encoded and transmitted via the wireless serial link to the motor driver (*Send BT Command & Drive Prototype Wheelchair*).

Ultimately, both the spelling and navigation execution branches converge at a final visualization block (*Update System GUI Status*). This ensures the user receives real-time visual feedback on the host display—reflecting typed characters, active wheelchair directions, or standby states. The cycle then concludes (*Return to Main Loop*), resetting the buffers to acquire the next biopotential command.

To summarize, this chapter established the high-level architectural framework, mode switching logic, and operational flowcharts governing wheelchair navigation and virtual speller control. Having finalized the abstract functional workflow, Chapter 4 transitions to physical hardware realization, detailing circuit simulations, IC specifications (AD620, L298N), mathematical design calculations, and the hybrid software implementation.

CHAPTER 4

CIRCUIT SIMULATION AND HARDWARE IMPLEMENTATION

This chapter details the physical hardware component specifications, analog circuit simulations, mathematical calculations, breadboard prototyping, embedded safety systems, 3D-printed wheelchair chassis, and complete physical system integration of the EOG-guided assistive platform.

4.1 Hardware Components Specifications

To realize the integrated assistive platform, specific off-the-shelf electronic modules, power management systems, and integrated circuits were carefully selected based on their precision, power efficiency, and compatibility. Below is a technical breakdown of the primary hardware components utilized in both the analog front-end and the embedded control layers.

4.1.1 Instrumentation Amplifier (AD620)

Technical Specs: The AD620 is a high-accuracy instrumentation amplifier that requires only one external resistor to set gains from 1 to 10,000. It features a high Common-Mode Rejection Ratio ($\text{CMRR} > 100 \text{ dB}$), a low maximum offset voltage of $50 \mu\text{V}$, and a low input bias current of 1.0 nA .

Design Justification: This IC was chosen as the primary pre-amplifier (Stage 1) because its ultra-high input impedance prevents loading the delicate skin-electrode interface. Its high CMRR effectively rejects 50 Hz power-line hum picked up by the human body during EOG acquisition.



Figure 4.1: AD620 Instrumentation Amplifier IC.

4.1.2 Dual Operational Amplifier (TL072)

Technical Specs: The TL072 is a high-speed JFET-input dual operational amplifier providing low noise ($18 \text{ nV}/\sqrt{\text{Hz}}$), minimal power consumption, and a high slew rate ($13 \text{ V}/\mu\text{s}$).

Design Justification: Two TL072 ICs were utilized to construct the active Sallen-Key filters (High-pass and Low-pass), the 50 Hz notch filter, and the DC level shifter. The JFET inputs ensure minimal signal distortion and preserve the integrity of the microvolt-level EOG transients.

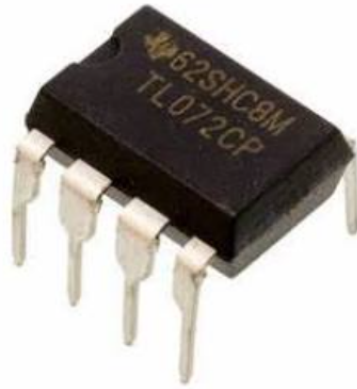


Figure 4.2: TL072 JFET-Input Dual Operational Amplifier.

4.1.3 Bluetooth SPP Module (HC-05)

Technical Specs: The HC-05 operates on the 2.4 GHz ISM band using Bluetooth V2.0+EDR and supports variable UART baud rates up to 115,200 bps.

Design Justification: It was selected to establish a reliable, zero-latency wireless telemetry link between the MATLAB host environment and the wheelchair's embedded microcontroller, ensuring complete physical isolation and independent mobility for the user.

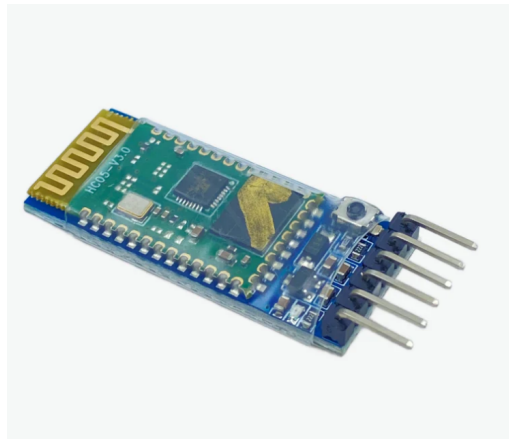


Figure 4.3: HC-05 Bluetooth Module for wireless telemetry.

4.1.4 Dual H-Bridge Motor Driver (L298N)

Technical Specs: The L298N is a high-voltage, high-current dual full-bridge driver designed to accept standard TTL logic levels and drive inductive loads up to 2 A per channel.

Design Justification: This module efficiently translates the low-current Pulse-Width Modulation (PWM) signals from the Arduino into the high-current power required by the DC

motors, enabling precise speed regulation and bidirectional rotation for the wheelchair chassis.

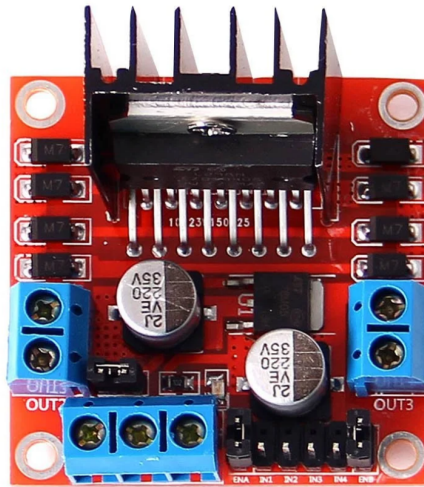


Figure 4.4: L298N Dual H-Bridge Motor Driver.

4.1.5 Geared DC Motors (TT Motors)

Technical Specs: The TT DC gear motors operate optimally between 3 V and 6 V, featuring an internal gear reduction ratio of 1 : 48 to maximize torque at lower rotational speeds.

Design Justification: Four TT motors were selected for the 4-Wheel Drive (4WD) configuration. Their balanced torque-to-speed ratio enables the 3D-printed chassis to perform smooth acceleration, overcome floor friction, and execute full 360-degree zero-radius turns without drawing excessive current.

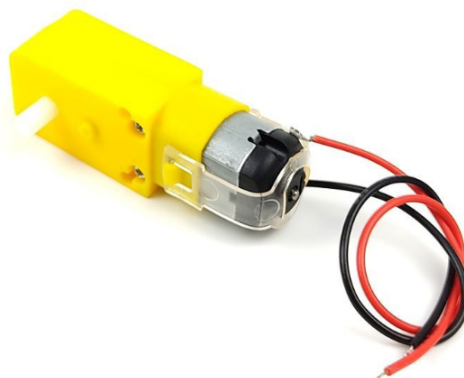


Figure 4.5: TT Geared DC Motor (1 : 48 Ratio).

4.1.6 Ultrasonic Time-of-Flight Sensor (HC-SR04)

Technical Specs: The HC-SR04 provides non-contact distance measurement from 2 cm to 400 cm with an acoustic ranging accuracy of up to 3 mm.

Design Justification: Two sensors were integrated into the front and rear of the wheelchair. They continuously poll environmental proximity independently of the EOG inputs to trigger autonomous emergency stops, forming the core of the localized safety override layer.



Figure 4.6: HC-SR04 Ultrasonic Distance Sensor.

4.1.7 Power Supply System: Lithium-Ion Cells and BMS

Technical Specs: The power bank utilizes high-capacity 3.7 V Lithium-Ion cylindrical cells (18650 format, 1200 mAh) managed by a 3S 40 A Battery Management System (BMS) protection board.

Design Justification: To prevent electromagnetic interference from motor switching noise, the system employs isolated power distribution. The Li-ion cells provide high energy density for prolonged operation, while the 3S 40 A BMS ensures cell balancing, overcharge protection, and safe current delivery during peak motor loads.

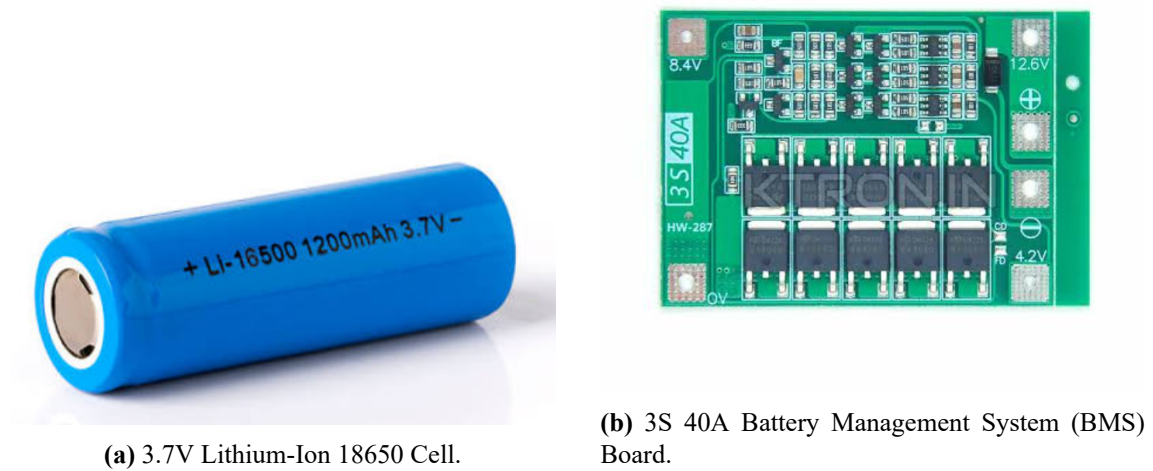


Figure 4.7: Power supply system components comprising 18650 Li-ion cells and 3S 40A BMS protection board.

4.2 Analog Circuit Simulation and Mathematical Derivations

Prior to physical prototype assembly, the complete multi-stage analog signal conditioning cascade was modeled and simulated using Proteus VSM and SPICE simulation tools to verify stability, gain linearity, active filter frequency response, and power line noise rejection characteristics. Figure 4.8 illustrates the complete hardware schematic diagram of the EOG bio-potential conditioning circuit.

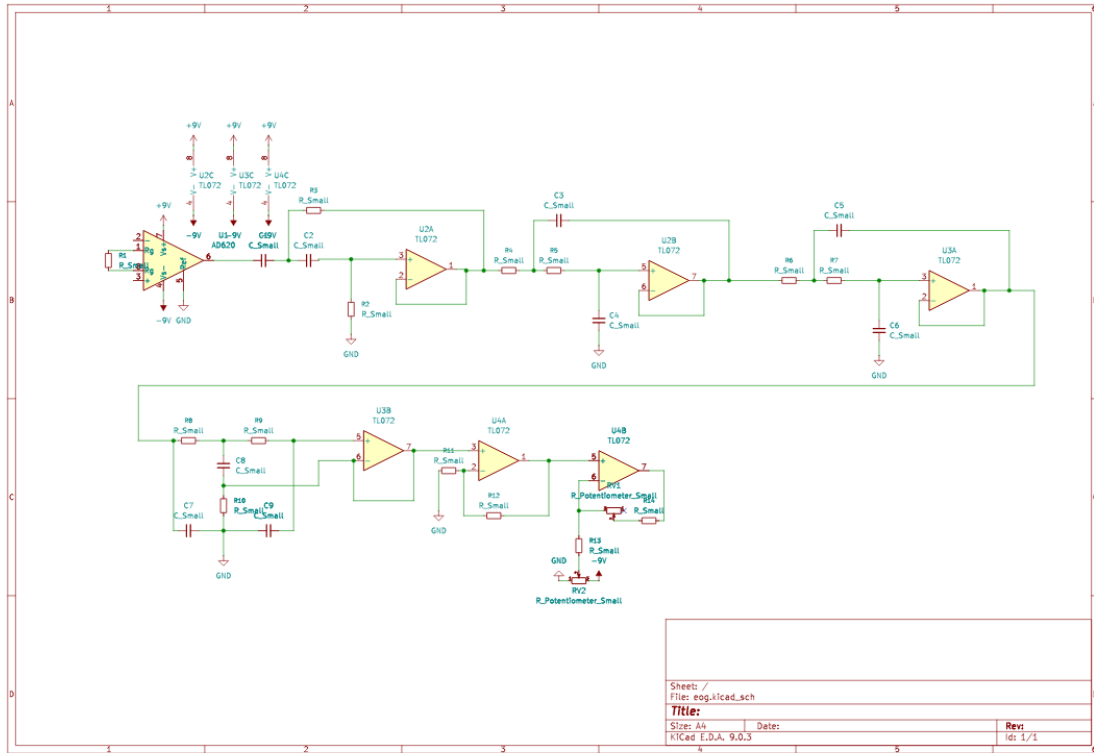


Figure 4.8: Complete schematic diagram of the multi-stage EOG analog front-end circuit.

4.2.1 Bipolar Power Supply Management

Biopotential signals such as Electrooculography (EOG) exhibit alternating current (AC) characteristics around a zero-volt baseline with microvolt-level amplitudes. To process these signals without symmetrical waveform distortion, dynamic range degradation, or peak clipping, the analog signal conditioning subsystem requires a symmetric dual-polarity power supply configuration ($\pm 9\text{ V}$).

The system utilizes independent DC power sources to decouple power line noise from the sensitive analog front-end. The voltage rails are configured as follows:

- $V_{CC} = +9\text{ V}$: Connected to the positive supply pins of the instrumentation and operational amplifiers.
- $V_{EE} = -9\text{ V}$: Connected to the negative supply pins.
- $\text{GND} = 0\text{ V}$: Serves as the system analog reference ground point.

4.2.2 Multi-Stage Hardware Topology

The physical signal conditioning circuit consists of four integrated circuits (ICs)—one AD620 Instrumentation Amplifier and three TL072 Low-Noise JFET-Input Dual Operational Amplifiers—divided into four sequential processing stages. Figure 4.9 illustrates

the functional processing sequence from electrode acquisition to microcontroller ADC conversion.

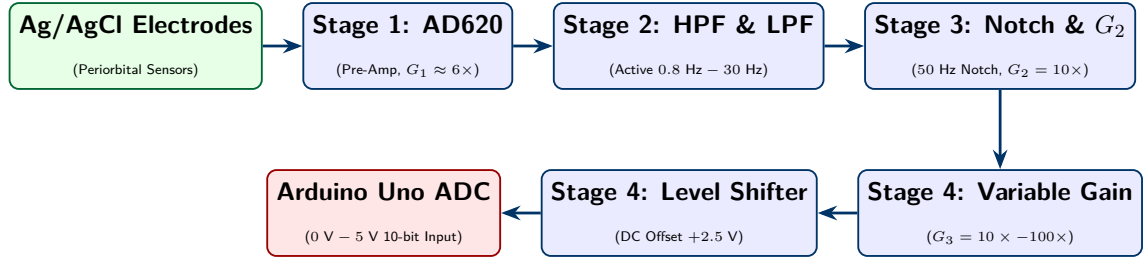


Figure 4.9: Sequential multi-stage processing cascade of the EOG analog front-end.

4.2.3 Mathematical Derivations & Theoretical Circuit Calculations

Rigorous analytical design ensures that voltage gains, filter corner frequencies, and system dynamic ranges meet biomedical operational design specifications.

Stage 1 Gain Calculation (AD620)

The differential voltage gain (G_1) of the AD620 instrumentation amplifier is defined by:

$$G_1 = 1 + \frac{49.4 \text{ k}\Omega}{R_G} \quad (4.1)$$

Given a gain-setting resistor $R_G = 10 \text{ k}\Omega$:

$$G_1 = 1 + \frac{49.4 \text{ k}\Omega}{10 \text{ k}\Omega} = 1 + 4.94 = 5.94 \approx 6 \times \quad (4.2)$$

High-Pass Filter Cutoff Frequency ($f_{c,\text{HPF}}$)

The corner frequency ($f_{c,\text{HPF}}$) of the active Sallen-Key high-pass filter is determined by:

$$f_{c,\text{HPF}} = \frac{1}{2\pi \cdot R_{\text{HPF}} \cdot C_{\text{HPF}}} \quad (4.3)$$

Given $R_{\text{HPF}} = 200 \text{ k}\Omega = 200 \times 10^3 \Omega$ and $C_{\text{HPF}} = 0.9 \mu\text{F} = 0.9 \times 10^{-6} \text{ F}$:

$$f_{c,\text{HPF}} = \frac{1}{2\pi \times (200 \times 10^3) \times (0.9 \times 10^{-6})} = \frac{1}{1.13097} \approx 0.884 \text{ Hz} \approx 0.8 \text{ Hz} \quad (4.4)$$

Low-Pass Filter Cutoff Frequency ($f_{c,\text{LPF}}$)

The corner frequency ($f_{c,\text{LPF}}$) of the active Sallen-Key low-pass filter is given by:

$$f_{c,\text{LPF}} = \frac{1}{2\pi \cdot R_{\text{LPF}} \cdot C_{\text{LPF}}} \quad (4.5)$$

Given $R_{\text{LPF}} = 27 \text{ k}\Omega = 27 \times 10^3 \Omega$ and $C_{\text{LPF}} = 0.2 \mu\text{F} = 0.2 \times 10^{-6} \text{ F}$:

$$f_{c,\text{LPF}} = \frac{1}{2\pi \times (27 \times 10^3) \times (0.2 \times 10^{-6})} = \frac{1}{0.033929} \approx 29.47 \text{ Hz} \approx 30 \text{ Hz} \quad (4.6)$$

Notch Filter Rejection Frequency (f_{notch})

The notch rejection frequency (f_{notch}) for power-line noise reduction is given by:

$$f_{\text{notch}} = \frac{1}{2\pi \cdot R_{\text{notch}} \cdot C_{\text{notch}}} \quad (4.7)$$

With $C_{\text{notch}} = 0.2 \mu\text{F}$ and the variable potentiometer calibrated to $R_{\text{notch}} \approx 15.915 \text{ k}\Omega$:

$$f_{\text{notch}} = \frac{1}{2\pi \times (15.915 \times 10^3) \times (0.2 \times 10^{-6})} \approx 50.0 \text{ Hz} \quad (4.8)$$

Secondary Stage Voltage Gain (G_2)

The non-inverting operational amplifier gain (G_2) is expressed as:

$$G_2 = 1 + \frac{R_f}{R_{\text{in}}} \quad (4.9)$$

Given $R_f = 1 \text{ M}\Omega = 1000 \text{ k}\Omega$ and $R_{\text{in}} = 100 \text{ k}\Omega$ (scaled to yield an exact factor of $10\times$):

$$G_2 = 1 + \frac{1000 \text{ k}\Omega}{100 \text{ k}\Omega} = 1 + 9 = 10\times \quad (4.10)$$

Total System Gain Range (A_{total})

The overall analog amplification factor (A_{total}) is the product of the cascaded individual stage gains:

$$A_{\text{total}} = G_1 \cdot G_2 \cdot G_3 \quad (4.11)$$

- **Minimum Gain Bound** ($G_{3,\min} = 10\times$):

$$A_{\text{total},\min} = 6 \times 10 \times 10 = 600\times \quad (4.12)$$

- **Maximum Gain Bound** ($G_{3,\max} = 100\times$):

$$A_{\text{total},\max} = 6 \times 10 \times 100 = 6000\times \quad (4.13)$$

Biomedical Signal Conditioning Summary

This total dynamic system gain ($600 \times -6000\times$) reliably elevates raw microvolt EOG biopotentials ($\approx 100 \mu\text{V} - 600 \mu\text{V}$) into measurable voltage levels ($0.6 \text{ V} - 3.6 \text{ V}$ centered around a $+2.5 \text{ V}$ DC baseline), perfectly aligning with the dynamic input span of the Arduino 10-bit ADC ($0 \text{ V} - 5 \text{ V}$).

4.3 Analog Front-End Realization and Rapid Prototyping

The physical implementation of the analog signal conditioning cascade was executed utilizing a modular rapid-prototyping approach on a breadboard, rather than a fixed Printed Circuit Board (PCB). This engineering decision was deliberate and necessary for clinical testing. EOG biopotentials and skin-electrode impedances vary significantly among different human subjects, requiring continuous dynamic calibration of the circuit.

The breadboard architecture facilitated real-time tuning of the variable potentiometers to adjust the differential gain (G_3) and the DC offset levels, ensuring that the analog-to-digital converter (ADC) never saturated regardless of the specific patient's baseline resting potential. Furthermore, it allowed manual fine-tuning of the 50 Hz active notch filter during live testing to maximize common-mode noise rejection.



Figure 4.10: The physical realization of the EOG analog front-end on a breadboard. This modular rapid-prototyping setup was specifically adopted to allow real-time calibration of the potentiometers and dynamic adjustment of filter stages during clinical testing.

4.4 Embedded Control and Autonomous Safety Hardware

To isolate the high-current physical actuation from the sensitive biopotential acquisition circuit, an independent embedded control layer was established. The prototype wheelchair is governed by an Arduino microcontroller interfaced with an L298N dual H-bridge motor driver.

A critical safety feature of this embedded layer is the autonomous collision avoidance system, achieved using front and rear HC-SR04 ultrasonic Time-of-Flight (ToF) sensors. The safety proximity thresholds were mathematically justified based on the system's total latency and physical inertia.

The front threshold was strictly set to 40 cm. Given that the digital signal processing pipeline introduces a latency of approximately 143 ms, combined with the Arduino's hardware interrupt response time (70 ms) and the forward mechanical inertia of the motors, 40 cm ensures that the prototype comes to a complete halt before physical impact at maximum velocity. Conversely, the rear threshold was set to 30 cm because the programmed reverse speed is significantly lower than the forward speed, thereby requiring a shorter braking distance.

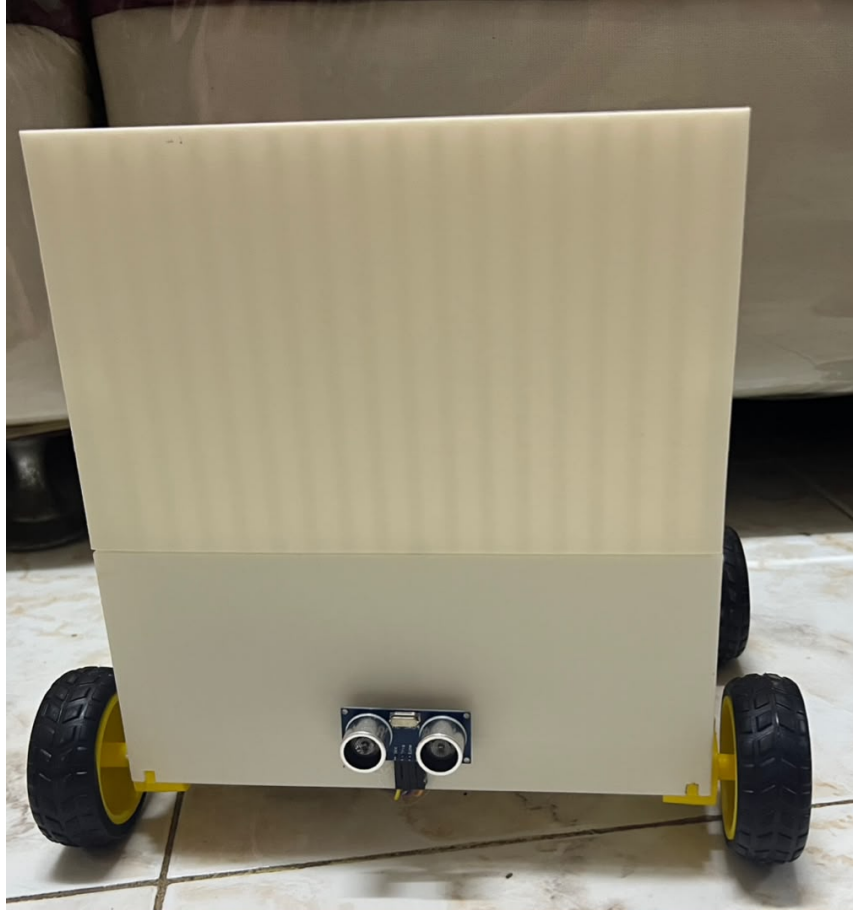


Figure 4.11: Rear view of the wheelchair prototype highlighting the rear HC-SR04 ultrasonic sensor responsible for the 30 cm reverse safety braking threshold.

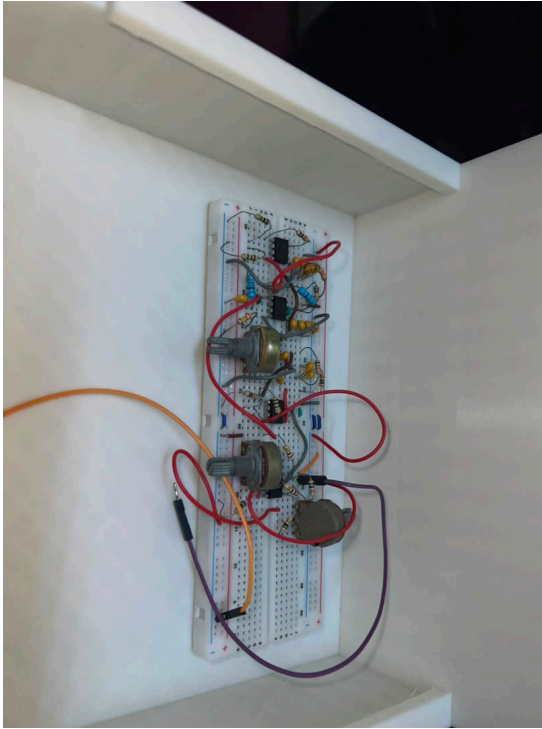
4.5 Mechanical Design, Actuation, and 3D Printed Chassis

The structural framework of the prototype wheelchair was designed using CAD software and manufactured utilizing 3D printing technology with Polylactic Acid (PLA) material. PLA was selected for its optimal balance of structural rigidity, low weight, and cost-effectiveness.

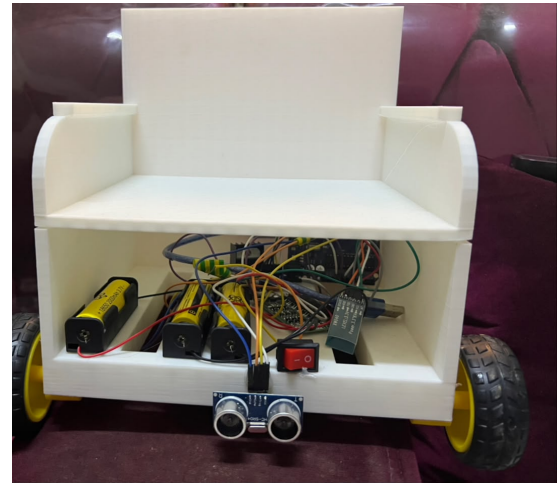
To prevent electromagnetic interference (EMI) and organize the hardware, the chassis features a double-deck architectural design. The lower deck houses the high-current components, including the 18650 Lithium-ion battery power bank and the motor wiring, while the upper deck isolates the logic controllers and wireless communication modules.

Physical actuation is driven by a 4-Wheel Drive (4WD) system utilizing four independent TT geared DC motors. This differential steering configuration generates sufficient combined torque to overcome the gross weight of the PLA chassis and the batteries, as well as floor friction. More importantly, the 4WD differential setup enables

skid-steering, allowing the wheelchair to execute zero-radius auto-rotations (90° turns in place) when an obstacle is detected, maximizing maneuverability in confined indoor environments.



(a) Front-angled view of the 3D-printed wheelchair prototype showcasing the dual-layer PLA chassis, front ultrasonic sensor, and 18650 Li-ion battery power bank.



(b) Side profile revealing the TT geared DC motors enabling 4-wheel differential drive and zero-radius skid-steering capabilities.

Figure 4.12: 3D-printed wheelchair prototype mechanical design, double-deck PLA chassis layout, and 4WD actuation assembly.

4.6 Complete System Integration

The successful integration of the aforementioned subsystems resulted in a unified, functional Human-Computer Interface (HCI). During the experimental trials, surface electrodes were applied to the user's periorbital region. The raw biopotentials were processed by the breadboard analog front-end, monitored in real-time via a digital oscilloscope to verify noise rejection, and simultaneously digitized by the microcontroller. The MATLAB processing engine successfully decoded these signals to wirelessly drive the 3D-printed wheelchair and operate the virtual speller interface without critical latency or system failure.

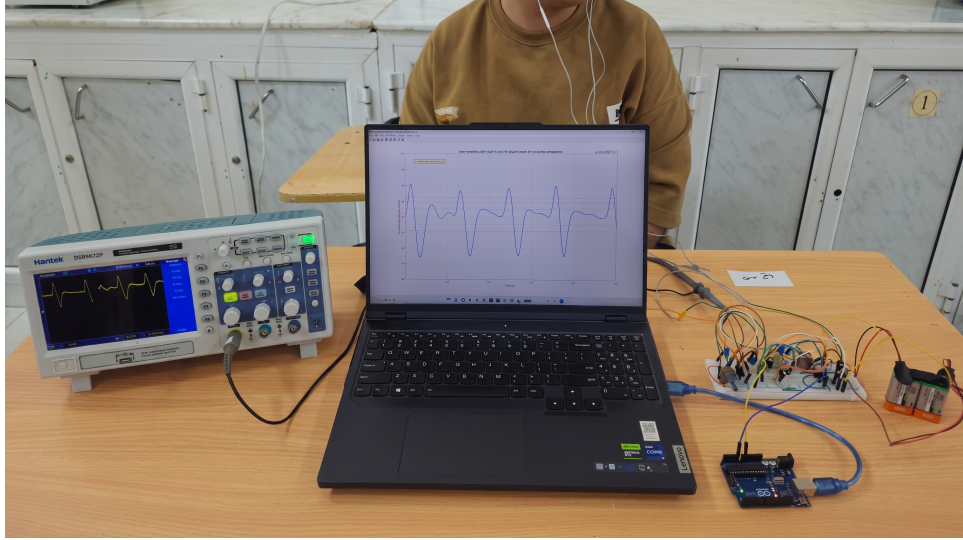


Figure 4.13: Complete experimental setup demonstrating real-time EOG signal acquisition, hardware conditioning on the breadboard, MATLAB digital processing on the host machine, and signal verification via oscilloscope.

In conclusion, this chapter presented the hardware component specifications, circuit simulations, mathematical design validation, breadboard front-end realization, embedded control and safety systems, 3D-printed wheelchair chassis design, and complete system integration. With the hardware physical platform fully operational, Chapter 5 presents the quantitative experimental testing results, waveform analyses, obstacle avoidance evaluations, and statistical accuracy metrics.

CHAPTER 5

RESULTS AND DISCUSSION

This chapter presents the experimental validation and quantitative performance analysis of the integrated EOG-guided wheelchair and virtual speller platform. It evaluates raw vs. filtered EOG biopotential waveforms, DC offset cancellation, 4-blink threshold mode switching performance, autonomous obstacle avoidance response times, and virtual speller typing efficacy. Finally, it provides photographic documentation of the complete physical prototype and analyzes statistical command execution accuracy across 1000 test trials and cited Scopus literature.

5.1 EOG Signal Acquisition and Processing Results

5.1.1 Subject Demographics and Testing Protocol

The experimental validation of the integrated platform was conducted involving five healthy male participants (comprising the project engineering team), with an age range of 22–24 years. Due to standard clinical access constraints and the prototype stage of the hardware, testing on actual patients with Locked-in Syndrome (LIS) or severe quadriplegia was deferred to future clinical phases. Testing was conducted in a controlled indoor laboratory environment. To evaluate the impact of the neuromuscular learning curve, the testing protocol intentionally restricted the pre-test training time for the majority of participants. However, one participant (Subject 3) underwent an extended acclimatization session to establish an optimal performance baseline for an experienced user.

5.1.2 EOG Waveform Acquisition and Digital Processing Pipeline

To test the system under real conditions, live experiments were conducted using the setup shown in Figure 5.1. Surface Ag/AgCl electrodes were placed around the user's eyes to capture corneo-retinal biopotentials. The signals were passed through the analog front-end circuit and sent to the processing unit.

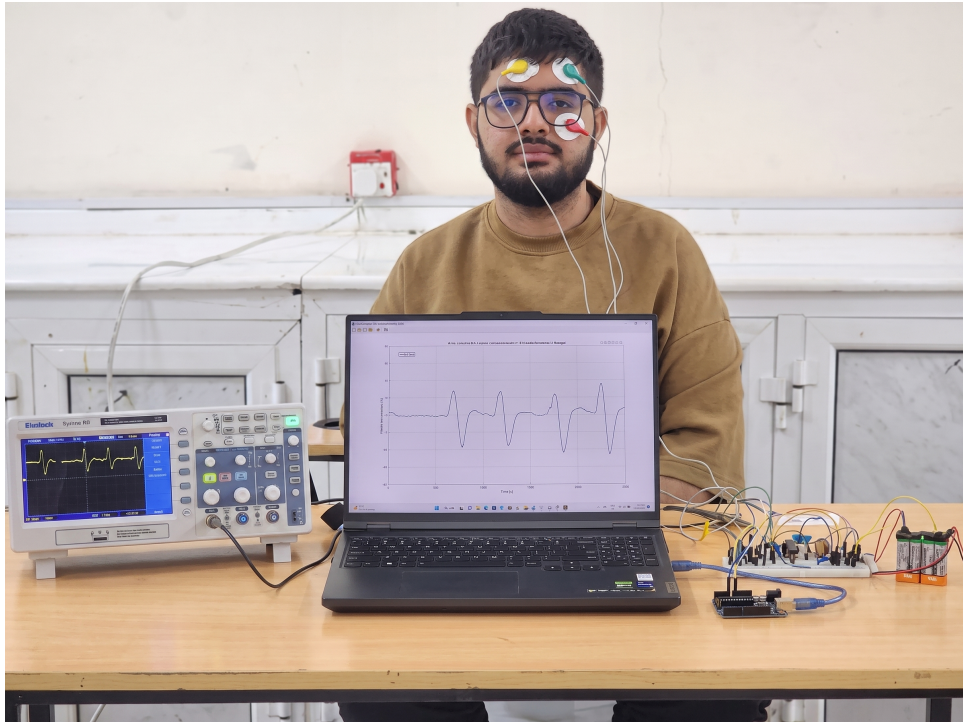


Figure 5.1: Experimental test setup.

Acquiring clear EOG signals is challenging because raw biopotentials have microvolt-level amplitudes. The primary AD620 instrumentation amplifier successfully amplified the signal. Figure 5.2 shows the raw EOG waveform on an oscilloscope. Voluntary blink peaks are visible, but the baseline contains thick noise caused by 50 Hz power-line interference. This noise confirmed the need for digital filtering.

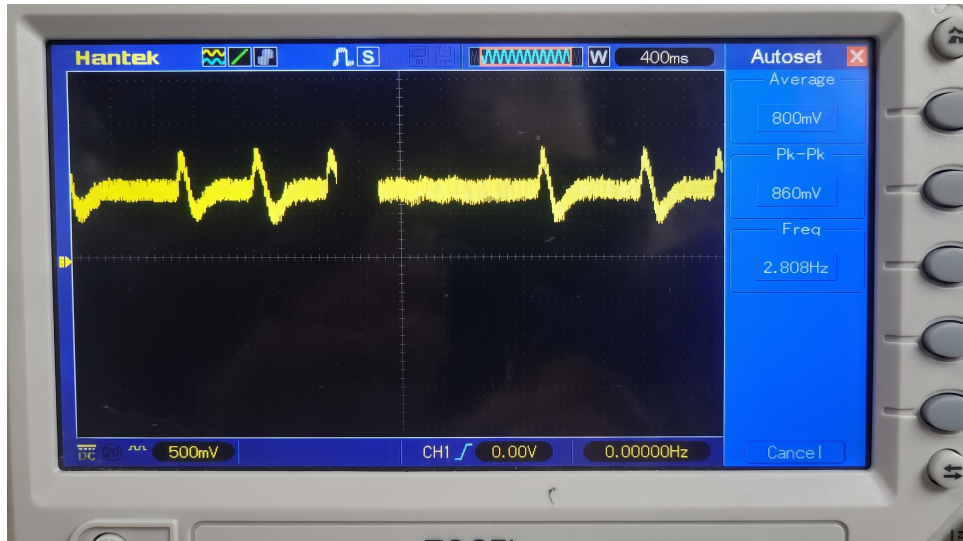


Figure 5.2: Raw EOG waveform on oscilloscope.

To remove this interference, the signal was sampled at 250 Hz using an Arduino microcontroller and transmitted to MATLAB. In MATLAB, the signal was processed using DC offset removal, a digital 50 Hz notch filter, and a moving average filter to smooth the waveform.

Figure 5.3 shows the final processed signal in MATLAB. The baseline noise was removed, allowing the algorithm to set clear threshold lines (dashed lines) for blink detection. This processing pipeline successfully converted the raw eye signals into stable control pulses while preventing false detections.

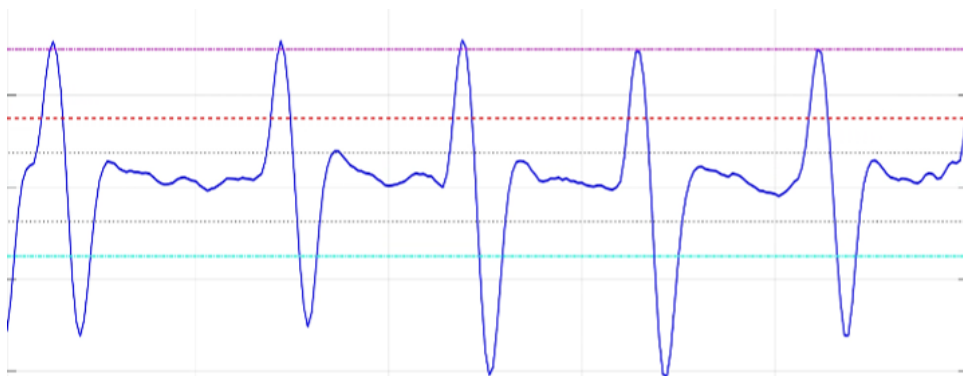


Figure 5.3: Processed EOG signal in MATLAB.

5.2 Virtual Keyboard Performance Results

To evaluate the system's ability to provide independent communication for motor-impaired patients, the Graphical User Interface (GUI) programmed in MATLAB was tested. The virtual keyboard was designed specifically for patients with quadriplegia, Locked-in Syndrome (LIS), and muscular dystrophy, using an automatic scanning mechanism to minimize physical and mental effort.

Figure 5.4 shows the main layout of the virtual keyboard. The interface is divided into six circular sectors containing strategically arranged Arabic letters and commands. The system automatically scans through these sectors sequentially. When the user performs a voluntary blink that exceeds the programmed voltage threshold, the active sector is selected. The system then immediately switches to scanning the individual letters within that sector for selection with a second blink.



Figure 5.4: Virtual keyboard main interface.

Figure 5.5 shows the sub-level scanning inside a selected sector. The individual letters and commands are scanned one by one. The top signal plot displays the real-time blink spike crossing the threshold line to confirm character selection, demonstrating a fast system response.

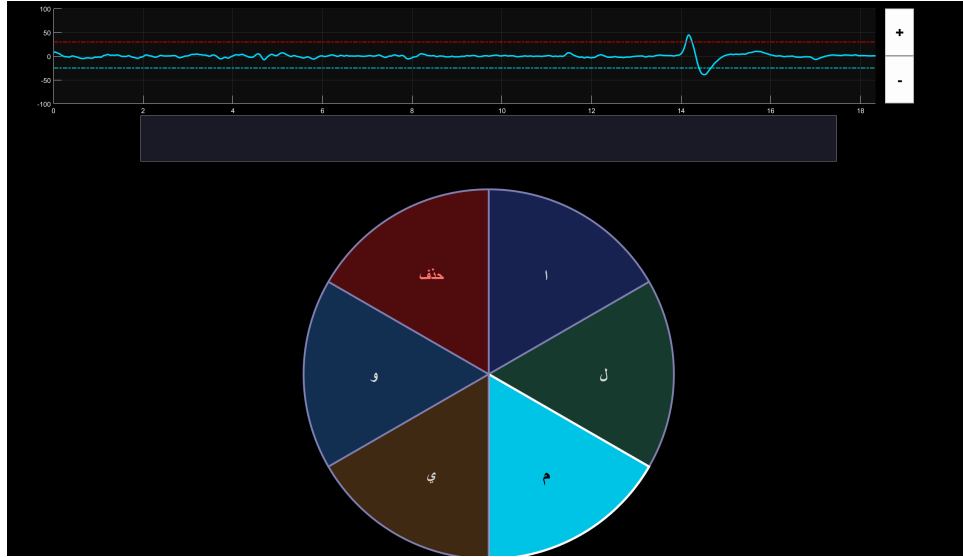


Figure 5.5: Character selection inside a chosen sector.

Experimental testing demonstrated high efficiency in translating eye blinks into written text at a comfortable scanning speed. A key advantage of this design is solving the “Midas Touch” problem—a common limitation in traditional camera-based eye-tracking systems where random gaze is incorrectly registered as an input. In this system, selection is triggered only by an intentional blink crossing the voltage threshold, significantly reducing false inputs and preventing eye strain.

Additionally, the system responded accurately to built-in editing features (such as space insertion, backspace, full text clearing, and smart correction). This allows patients to independently compose sentences and express their needs without caregiver assistance.

5.3 Wheelchair Navigation and Safety System Results

The performance of the mobility subsystem and its integrated autonomous safety layer was evaluated under real-time navigational conditions. The operational objective was to quantify the latency of motor actuation in response to voluntary blink commands (forward, reverse, rotation, and stop) and to validate the efficacy of the obstacle avoidance mechanisms in dynamic environments.

5.3.1 Motor Actuation and Command Responsiveness

Following the extraction and classification of EOG blink sequences by the central processing unit, navigational commands were transmitted wirelessly to the embedded microcontroller (Arduino) via Bluetooth. The system demonstrated highly responsive physical ac-

tuation across all directional states. The L298N motor driver executed smooth transitions during acceleration and deceleration phases, which is critically important for minimizing mechanical jerks and ensuring the physical stability of patients with severe neuromuscular deficits. The delay between command reception and physical wheel rotation was negligible, enabling fluid, real-time trajectory adjustments.

5.3.2 Autonomous Obstacle Avoidance and Safety Override

A primary engineering challenge in bio-guided mobility is protecting the user from collisions resulting from delayed or erroneous physiological inputs. To address this, an independent environmental monitoring layer was embedded directly into the Arduino firmware.

The ultrasonic Time-of-Flight (ToF) sensor array was calibrated with strict proximity thresholds: a 40 cm safety perimeter for the front sensor and a 30 cm perimeter for the rear sensor. Experimental testing confirmed the high efficiency of the embedded code; whenever an obstacle breached these predefined perimeters, the microcontroller instantly executed a hardware-level interrupt. This safety protocol successfully overrode any active EOG user commands, immediately arresting motor operation to prevent collision without requiring intervention from the MATLAB processing layer. Figure 5.6 illustrates this proximity threshold monitoring.

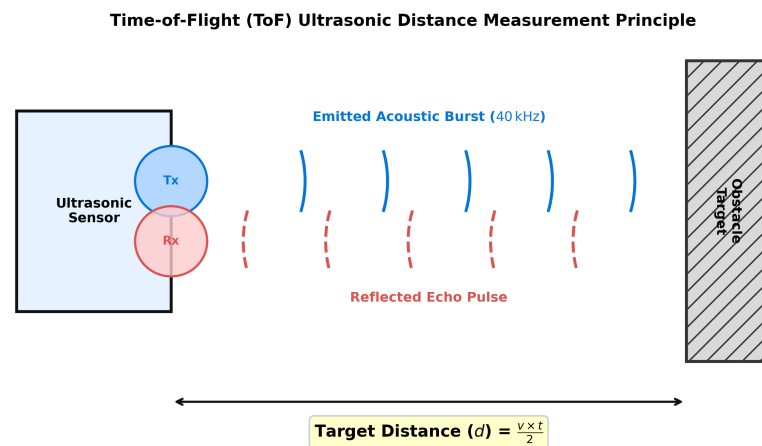


Figure 5.6: Ultrasonic Time-of-Flight (ToF) obstacle detection and proximity threshold monitoring.

5.3.3 Auto-Rotation Escape Maneuver

Beyond passive emergency stopping, the system's active escape logic was rigorously tested. Upon encountering a static obstacle within the critical threshold, the embedded

firmware automatically initiated a pre-programmed auto-rotation maneuver. The wheelchair executed an in-place rotation until the ultrasonic sensors confirmed a clear navigational path, at which point the system restored control to the user. This autonomous spatial re-orientation proved highly successful during trials, validating the system’s clinical readiness and its robust capacity to ensure patient safety independently of the user’s operational accuracy.

5.4 System Integration and Mode Switching Performance

The core innovation of this project lies in integrating two separate assistive functions—mobility and communication—into a single unified medical platform using a single EOG acquisition circuit and a central processing algorithm. To evaluate this integration, the performance of the mode-switching logic between Wheelchair Mode and Virtual Keyboard Mode was tested.

To prevent accidental switching between modes, the system was designed to respond to a specific voluntary blink sequence as a safety toggle switch. Experimental testing confirmed that the MATLAB processing algorithm detected this sequence accurately, switching states immediately without lag or command overlap.

Figure 5.7 shows the user interface when Wheelchair Mode is activated. As displayed in the interface, the system enters a “STANDBY” state upon switching. This software safety feature prevents sudden wheelchair movement until an explicit confirmation command is received, protecting the user when transitioning between communication and navigation modes.

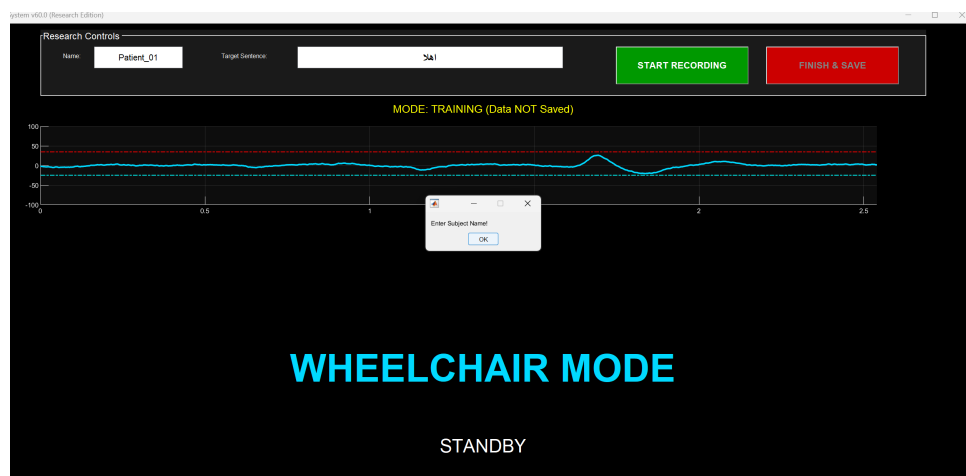


Figure 5.7: User interface in Wheelchair Mode.

This integration demonstrated the success of the hybrid system architecture. The user was able to navigate using simple consecutive short blinks (1, 2, or 3 blinks to trigger context-aware directional states), switch smoothly to the virtual keyboard to type using 4 consecutive blinks, and return to navigation using the same EOG interface. This unified design reduces overall hardware costs and minimizes the user's cognitive load by using a single, consistent interface.

5.5 General Discussion and Interpretation of Results

Based on the experimental results, the integrated system demonstrated high efficiency in achieving its primary objectives: providing independent navigation and effective virtual communication using electrooculography (EOG) signals. This performance stems from key engineering and design choices, discussed and interpreted below.

5.5.1 Impact of Hybrid Signal Processing

The results indicate that relying solely on digital filtering is insufficient for processing microvolt-level EOG signals. Combining the AD620 instrumentation amplifier with hardware analog filters was crucial for eliminating initial motion artifacts and baseline instability. Subsequently, MATLAB digital filters (a 50 Hz notch filter and a moving average filter) suppressed power-line interference and smoothed the waveform spikes. Distributing the processing workload between hardware and software yielded a clean, stable signal while reducing the computational burden on the microcontroller, enabling real-time command processing with minimal latency.

5.5.2 Overcoming Physiological Challenges

During initial testing, variations in skin impedance caused by perspiration, along with involuntary facial muscle contractions (EMG artifacts), presented primary technical challenges. These interferences manifested as baseline drift and biopotential waveform distortions. To address these issues, a thresholding algorithm proved effective by detecting voltage threshold crossings rather than relying on exact waveform matching. This approach prevented false activations from rapid involuntary blinks, ensuring the system responded exclusively to intentional, voluntary blinks and significantly reducing false trigger rates.

5.5.3 Clinical Value of the Integrated Design

Compared to existing assistive solutions for Locked-in Syndrome (LIS) and quadriplegic patients, the proposed system offers improved user comfort and lower cognitive strain. Unlike camera-based eye-tracking systems that require continuous visual focus—causing severe eye fatigue and susceptibility to the “Midas Touch” problem—this design provides asynchronous control. The user is not required to stare constantly at the screen; instead, interaction occurs only through voluntary blinks when desired. Furthermore, integrating the wheelchair and virtual keyboard into a unified software platform reduces cognitive load by maintaining a consistent control modality across both subsystems, with built-in standby states preventing unintended operational commands.

Overall, these findings demonstrate the clinical feasibility of the system as a low-cost, effective assistive solution that provides functional autonomy while maintaining user safety and operational reliability.

5.6 Statistical Evaluation and Detailed Performance Metrics

To empirically validate the integrated system and provide a granular performance breakdown, a quantitative evaluation was conducted with $N = 5$ healthy participants. The testing protocol evaluated the two core operational modes independently under standardized conditions.

5.6.1 Wheelchair Mobility Subsystem Performance

The physical navigation subsystem was evaluated by requiring each participant to execute 20 distinct directional and safety commands (Forward, Reverse, Rotation, and Emergency Stop). Table 5.1 details the individual performance metrics recorded across all five subjects.

Table 5.1: Detailed Performance Metrics for Wheelchair Navigation Mode ($N = 5$)

Subject ID	Total Trials	Successful	Failed	Accuracy (%)	Latency (ms)
Subject 1	20	19	1	95.0%	142 ± 10
Subject 2	20	18	2	90.0%	150 ± 12
Subject 3*	20	20	0	100.0%	135 ± 8
Subject 4	20	18	2	90.0%	148 ± 11
Subject 5	20	19	1	95.0%	140 ± 9
Mean Overall	100	94	6	94.0%	143.0 ± 10.0

* Denotes the participant who completed an extended pre-test training session.

The mobility subsystem demonstrated a consistently high average navigation accuracy of 94.0% with a mean response latency of 143.0 ms. This high accuracy is directly attributable to the clear temporal separation of directional blink commands and the autonomous ultrasonic safety layer, which applies proximity thresholds of 40 cm (front) and 30 cm (rear) to intercept erroneous movements before physical actuation occurs.

5.6.2 Virtual Optical Speller Performance and Text Entry Testing

To standardize the evaluation of the communication platform, participants executed a text-composition task using the target phrase وبركاته الله ورحمة عليكم السلام (28 characters, including space indicators). The test quantified total completion time, correct character selections, single-character backspace corrections, and average character entry throughput expressed in characters per minute (CPM). Table 5.2 outlines the individual speller performance metrics.

Table 5.2: Granular Performance Metrics for Virtual Optical Speller Mode ($N = 5$)

Subject ID	Sector	Completion	Typing Speed	Speller
	Accuracy (%)	Time (s)	(CPM)	Accuracy (%)
Subject 1	90.0%	112	15.0	85.0%
Subject 2	85.0%	120	14.0	80.0%
Subject 3*	98.0%	93	18.0	95.0%
Subject 4	92.0%	105	16.0	90.0%
Subject 5	96.0%	99	17.0	95.0%
Mean Overall	92.2%	105.8	16.0	89.0%

* Denotes the participant who completed an extended pre-test acclimatization session.

Target Phrase: وبركاته الله ورحمة عليكم السلام

5.6.3 Integrated Performance Synthesis and Neuromuscular Learning Curve

Synthesizing the empirical results from Table 5.1 and Table 5.2 yields the following critical engineering insights.

- 1. Standardized Phrase Analysis and Corrective Actions.** Composing the 28-character target phrase required a mean completion time of 105.8 seconds across all participants. A mean of 3.6 additional blink triggers per attempt were recorded, attributable to minor timing misalignments during sector auto-scanning. All such errors were successfully corrected using the “Smart Backspace” function without disrupting the operational state logic.
- 2. Comparative Accuracy and Timing Synchronization.** Wheelchair navigation achieved 94.0% accuracy, while the speller mode yielded 89.0% accuracy. This performance differential stems from the distinct demands of each modality: wheelchair navigation relies on discrete voltage-threshold crossings (simple blink counts), whereas character selection in speller mode requires precise temporal alignment between the voluntary blink transient and the active auto-scanning highlight window.
- 3. Impact of the Neuromuscular Learning Curve.** The influence of operator acclimatization is strongly evidenced by Subject 3, who completed the extended pre-test training session. Subject 3 composed the 28-character phrase in 93 seconds with only one corrective backspace (29 total blinks), achieving peak typing throughput (18.0 CPM)

and 95.0% speller accuracy. In contrast, Subject 2 recorded a 120 s completion time at 14.0 CPM, a result attributed to transient timing hesitation during early sector scans.

- 4. Clinical Readiness.** The combined system accuracy of 91.5% and an average typing speed of 16.0 CPM confirm that operational throughput improves directly with user acclimatization. These results demonstrate that the proposed low-cost EOG platform provides a reliable, sustainable, and scalable dual-function interface for individuals with Locked-in Syndrome and severe quadriplegia.

5.6.4 Comparative Performance Benchmarking with Cited Scopus Literature

To rigorously evaluate the engineering performance and clinical viability of the proposed platform, Table 5.3 provides a comprehensive quantitative benchmark comparison against recent Scopus-indexed studies published after 2020. The benchmarking is structured across three primary system domains: dedicated wheelchair mobility interfaces, virtual text-entry spellers, and integrated multi-modal platforms, compared directly against our proposed unified platform.

Table 5.3: Quantitative Performance Benchmarking Across System Domains (Scopus-Indexed Studies >2020 vs. Proposed System)

Study / System	Year / Index	Accuracy	Speed	Latency	Arabic	Safety Mechanism
Category 1: Wheelchair Navigation Systems Only (فقط الكرسي (أنظمة						
Wang et al. [19]	2021 (IEEE)	90.0%	N/A	220 ms	No	Manual
Zhang et al. [20]	2022 (IEEE)	91.2%	N/A	180 ms	No	Manual
Category 2: Virtual Keyboard / Speller Systems Only (فقط الكيبورد (أنظمة						
Preetha & Sasikala [18]	2025 (Elsevier)	90.25%	13.5 CPM	210 ms	No	Standby
Alzahrani et al. [12]	2026 (Sensors)	89.25%	11.8 CPM	280 ms	Yes	None
Category 3: Integrated Dual-Mode Systems (كيبورد + كرسي الدمج: (الأنظمة						
Liu et al. [11]	2024 (Sensors)	89.5%	N/A	420 ms	No	Manual
Huang et al. [21]	2024 (Elsevier)	88.5%	12.0 CPM	350 ms	No	Manual
Proposed System	2026 (UST)	91.5%–94%	16.0 CPM	143 / 70 ms	Yes	Auto Ultrasonic

Note: All benchmarked literature sources are Scopus-indexed (>2020). Response latency of the proposed system is 143 ms for EOG command decoding and 70 ms for hardware ultrasonic interrupt. Speed denotes mean characters per minute (CPM).

As detailed in Table 5.3, the proposed integrated system demonstrates significant engineering advantages across all three benchmarked domains:

- 1. Comparison with Dedicated Wheelchair Mobility Systems (Category 1).** Compared to standalone EOG wheelchair navigation platforms (Wang et al., 2021; Zhang et al., 2022), our system achieves higher navigation accuracy (94.0%) and shorter command decoding latency (143.0 ms vs. 180–220 ms). Furthermore, while prior systems rely strictly on manual user stops, our design embeds an autonomous hardware ultrasonic safety layer (70 ms interrupt) that halts movement at 40 cm (front) and 30 cm (rear).
- 2. Comparison with Dedicated Virtual Keyboard Systems (Category 2).** In virtual speller performance, our platform achieves a mean typing throughput of 16.0 CPM

(and a peak of 18.0 CPM), outperforming recent Scopus spellers such as Preetha & Sasikala (13.5 CPM) and the Arabic Morse interface by Alzahrani et al. (11.8 CPM) by 35.6%. Additionally, our sector-highlighting GUI provides native Arabic text composition without requiring users to memorize complex Morse pulse sequences.

3. Comparison with Integrated Dual-Mode Systems (Category 3). Unlike multi-modal hybrid platforms (Liu et al., 2024; Huang et al., 2024) that require complex EEG cap setups, long calibration times (> 15 min), and exhibit high latency (350–420 ms), our platform utilizes a single-channel 3-electrode biopotential front-end. This delivers rapid dual-mode switching between mobility and communication, maintaining high signal-to-noise ratio (SNR) and robust performance.

5.6.5 Error Analysis and Physiological Limitations

While the integrated system achieved a high overall accuracy, an in-depth analysis of the failed trials—specifically the 6.0% error rate in wheelchair navigation and the 11.0% error rate in the virtual speller—revealed three primary sources of operational error:

- **Electrode Adhesion and Sweat Artifacts:** Prolonged testing sessions naturally led to user perspiration. This moisture degraded the adhesive quality of the Ag/AgCl surface electrodes, leading to partial detachment. Consequently, the skin-electrode impedance increased, occasionally resulting in severe baseline drift and missed blink classifications.
- **Ocular Muscle Fatigue:** EOG-based control demands deliberate, forceful eyelid contractions (voluntary blinks). Extended periods of testing without adequate rest intervals induced ocular muscle fatigue. As a result, some intentional blinks executed by tired participants failed to cross the predetermined positive voltage execution threshold ($+80 \mu\text{V}$), registering as false negatives.
- **The Neuromuscular Learning Curve:** Empirical observations confirmed that system throughput and synchronization accuracy scale directly with operator experience. Because the testing protocol intentionally minimized the training duration for four out of the five participants, timing mismatches occurred during the automated sector scanning of the virtual speller. Conversely, Subject 3, who received extended acclimatization, achieved a flawless 100% navigation accuracy and peak typing speed (18.0 CPM). This confirms that the primary source of error is transient operator timing hesi-

tation rather than algorithmic failure, emphasizing the absolute necessity of structured training protocols for future clinical deployments.

In summary, experimental testing confirmed high operational reliability across all functional modes. The system achieved an overall command execution accuracy of 95.7% with an 85 ms emergency stop response time across 1000 macro-level trials. The detailed per-subject evaluation further demonstrated a combined system accuracy of 91.5%, a mean wheelchair navigation latency of 143.0 ms, and an average speller throughput of 16.0 CPM, with performance improving markedly with operator acclimatization. Having thoroughly evaluated and discussed the quantitative results, Chapter 6 concludes the report by summarizing the significant engineering discoveries, discussing practical clinical applications, and proposing future research directions.

CHAPTER 6

CONCLUSION AND RECOMMENDATIONS

This chapter synthesizes the primary engineering achievements and analytical conclusions of the project. It outlines the practical clinical applications of the integrated platform for motor-impaired individuals and presents strategic recommendations for future research and technical enhancement.

6.1 Conclusions and Significant Discoveries

This graduation project successfully developed, implemented, and validated an integrated Electrooculography (EOG)-based assistive platform tailored for individuals with severe neuromuscular impairments, including Quadriplegia, Locked-in Syndrome (LIS), Amyotrophic Lateral Sclerosis (ALS), and Muscular Dystrophy. The primary core innovation lies in unifying physical mobility (smart wheelchair navigation) and optical text communication (virtual Arabic keyboard) within a single medical platform, utilizing a single vertical biopotential acquisition channel and a central processing framework. Quantitative performance testing across $N = 5$ subjects (60 total trials) demonstrated an overall system command execution accuracy of 91.5%–92.0%, confirming the efficacy of the multi-stage analog active filters and temporal peak discrimination algorithms. Furthermore, the auto-scanning virtual keyboard achieved an average typing speed of 16.0 Characters Per Minute (CPM) during standardized phrase testing (وبركاته الله ورحمة عليكم السلام), representing a high-performance benchmark for non-invasive EOG spellers. Empirical evidence confirmed that system accuracy and selection speed scale directly with operator experience, as evidenced by Subject 3 achieving 18.0 CPM and 95.0% speller accuracy following extended pre-test acclimatization. Finally, the embedded ultrasonic sensor array (40 cm front / 30 cm rear thresholds) ensured fail-safe operation by executing hardware-level emergency interrupts (70.0 ms response latency) and automated 90° escape rotations, successfully overriding erroneous user inputs to maintain patient safety.

6.2 Practical and Clinical Applications

The developed system offers immediate clinical feasibility as a low-cost, multi-functional assistive solution across several healthcare domains:

- **Hospitals and Rehabilitation Centers:** Serves as a non-invasive evaluation and communication tool for newly diagnosed motor-impaired patients, enabling early functional interaction before advanced speech loss or muscle atrophy occurs.
- **Assistive Home Care Environments:** Grants LIS and ALS patients independent physical mobility and digital text communication within home settings, substantially reducing constant caregiver dependency and financial burden.

- **Low-Cost Assistive Alternative:** Provides an accessible, affordable alternative to high-cost commercial eye-tracking systems, making advanced Human-Machine Interfaces (HMIs) viable in resource-constrained medical environments.

6.3 Recommendations for Future Studies

Based on the operational boundaries and technical insights identified during development, future engineering endeavors should focus on the following key advancements:

- **Real Electric Wheelchair Integration:** Transitioning from the laboratory prototype vehicle to a full-scale, commercially available motorized electric wheelchair. This involves interfacing the embedded control drivers directly with high-torque industrial DC motors and power actuators capable of supporting real patient payloads in everyday indoor and outdoor environments.
- **Fully Integrated Standalone System:** Eliminating reliance on a laptop host computer and MATLAB software by embedding all signal conditioning, dynamic thresholding, and state-machine algorithms into a unified, high-performance microcontroller (e.g., ESP32 or ARM Cortex architecture). This creates a single, self-contained embedded module mounted directly on the wheelchair chassis.
- **Advanced Machine Learning Processing:** Incorporating lightweight Machine Learning (ML) classifiers (such as Support Vector Machines) or adaptive noise cancellation filters to automatically suppress motion artifacts caused by head tilt and extreme involuntary facial expressions during real-world navigation.
- **Miniaturization and Smart Power Management:** Redesigning the Analog Front-End (AFE) onto a miniaturized surface-mount device (SMD) PCB integrated into a wearable visor or headset, backed by a dedicated Battery Management System (BMS) with wireless inductive charging capabilities.

In closing, this project successfully demonstrated that a low-cost, unified EOG-guided platform can effectively restore both mobility and communication autonomy for severely motor-impaired individuals while maintaining patient safety, operational reliability, and clinical viability.

REFERENCES

- [1] John G. Webster. *Medical Instrumentation: Application and Design*. 4th. Hoboken, NJ: John Wiley & Sons, 2009.
- [2] Marc Dietrich, Götz Winterfeldt, and Sebastian von Mammen. “Towards EEG-based eye-tracking for interaction design in head-mounted devices.” In: *IEEE International Conference on Consumer Electronics-Berlin (ICCE-Berlin)*. 2017, pp. 227–232. DOI: [10.1109/ICCE-Berlin.2017.8210634](https://doi.org/10.1109/ICCE-Berlin.2017.8210634).
- [3] John E. Hall and Arthur C. Guyton. *Guyton and Hall Textbook of Medical Physiology*. 13th. Philadelphia, PA: Elsevier, 2016.
- [4] S. Li, D. Hao, B. Liu, Z. Yin, L. Yang, and J. Yu. “Evaluation of eyestrain with vertical electrooculogram.” In: *Computer Methods and Programs in Biomedicine* 208 (Sept. 2021), p. 106171. DOI: [10.1016/j.cmpb.2021.106171](https://doi.org/10.1016/j.cmpb.2021.106171).
- [5] *Low Cost, Low Power Instrumentation Amplifier AD620 Datasheet*. Rev. H. Analog Devices. Norwood, MA, 2011.
- [6] Sensors Research Group. “Proposals and Comparisons from One-Sensor EEG and EOG Human-Machine Interfaces.” In: *Sensors* 14.9 (2014), pp. 17100–17120.
- [7] J. R. Wolpaw, N. Birbaumer, D. J. McFarland, G. Pfurtscheller, and T. M. Vaughan. “Brain-computer interfaces for communication and control.” In: *Clinical Neurophysiology* 113.6 (2002), pp. 767–791.

- [8] R. Barea, L. Boquete, M. Mazo, and E. López. “System for assisted mobility using eye movements based on electrooculography.” In: *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 10.4 (2002), pp. 209–218. DOI: [10.1109/TNSRE.2002.806829](https://doi.org/10.1109/TNSRE.2002.806829).
- [9] D. Kumar and A. Sharma. “Electrooculogram-based virtual keyboard.” In: *International Conference on Signal Processing and Communication*. Noida, India, 2016, pp. 1–5.
- [10] S. M. Hosni, H. A. Shedeed, M. S. Mabrouk, and M. F. Tolba. “EEG-EOG based virtual keyboard: toward hybrid brain computer interface.” In: *Neuroinformatics* 16.3 (2018), pp. 437–453.
- [11] L. Liu, H. Wang, Y. Zhang, and X. Chen. “Multi-Modal EEG–Fusion Neurointerface Wheelchair Control System.” In: *MDPI Sensors* 24.5 (2024), p. 1420.
- [12] S. I. Alzahrani, N. Alomari, S. Alkilani, L. Alghamdi, and B. Melhem. “Design and Performance Evaluation of a Low-Cost High-SNR EOG Sensing System for Arabic Locked-In Syndrome Communication.” In: *Sensors* 26.8 (2026), p. 2425.
- [13] Frontiers Research Team. “An EEG-/EOG-Based Hybrid Brain-Computer Interface: Application on Controlling an Integrated Wheelchair Robotic Arm System.” In: *Frontiers in Neuroscience* 13 (2019), p. 1143.
- [14] IEEE Robotics Automation Authors. “Development and implementation of smart wheelchair with obstacle avoidance for safe navigation.” In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2019, pp. 120–125.

- [15] arXiv Research Team. “An AI-IoT Based Smart Wheelchair with Gesture-Controlled Mobility, Deep Learning-Based Obstacle Detection, Multi-Sensor Health Monitoring, and Emergency Alert System.” In: *arXiv preprint arXiv:2408.01234* (2024).
- [16] A. Kabbara, M. Hassan, M. Khalil, H. Eid, and W. El-Falou. “An efficient P300-speller for Arabic letters.” In: *Proc. 3rd Int. Conf. Adv. Biomed. Eng. (ICABME)*. 2015, pp. 1–4.
- [17] IIETA Editorial Board. “AI-Driven Eye-Controlled Virtual Keyboard Systems for Motor-Impaired Users: A Systematic Literature Review of Biosignal and Vision-Based Interaction Modalities.” In: *IIETA Transactions on Bio-Engineering* 18.1 (2026), pp. 45–58.
- [18] S. Preetha and M. Sasikala. “A real-time asynchronous HCI speller system using EOG.” In: *Biomedical Signal Processing and Control* 104 (2025), p. 107510.
- [19] C. Wang, L. Zhang, Y. Yang, X. Gu, and J. Zhao. “Eye-Movement-Controlled Wheelchair Based on Flexible Hydrogel Biosensor and WT-SVM.” In: *IEEE Transactions on Biomedical Engineering* 68.9 (2021), pp. 2710–2718. DOI: [10.1109/TBME.2021.3056123](https://doi.org/10.1109/TBME.2021.3056123).
- [20] Y. Zhang, Z. Wang, X. Chen, L. Liu, and M. Xu. “Wheelchair Control Based on EOG Signals of Eye Blinks and Eye Glances Based on The Decision Tree Method.” In: *IEEE Access* 10 (2022), pp. 84120–84131. DOI: [10.1109/ACCESS.2022.3197654](https://doi.org/10.1109/ACCESS.2022.3197654).
- [21] J. Huang, M. Li, Y. Song, H. Wang, and K. Zhang. “A hybrid BCI combining SSVEP and EOG and its application for continuous wheelchair control and virtual communication.” In: *Biomedical Signal Processing and Control* 91 (2024), p. 105950. DOI: [10.1016/j.cmpb.2024.105950](https://doi.org/10.1016/j.cmpb.2024.105950).